The Economic Costs of Cyber Risk

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Abstract

How does rising cyber risk affect the macroeconomy, and how should policymakers respond? We develop a general equilibrium model to quantify the economic costs of cyber risk. The model features strategic interactions between heterogeneous firms investing in cybersecurity and attackers targeting firms. Cyberattacks generate a negative externality by reducing aggregate productivity. We discipline the model using new firm-level evidence that the share of workers employed in cybersecurity occupations rises steeply with firm size. Because larger firms are more attractive targets but also invest more in protection, the model generates an inverse-U relationship between firm size and attack risk, consistent with the data. Introducing cyber risk reduces firm entry by 3.6 percent, aggregate productivity by 0.6 percent, and total output by 1.8 percent. These effects arise from general equilibrium adjustments in entry, firm size, and spillovers that are absent in typical partial-equilibrium analyses. Policy responses differ sharply: appropriately designed subsidies and minimum cybersecurity requirements can raise aggregate output, while bailouts reduce it.

Keywords: Cyber Risk, Search and Matching, Firm Dynamics, Macroeconomic Policy

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1 Introduction

Over the past two decades, cyber risk has evolved from a narrow IT concern into a widespread threat that firms across the economy confront daily. As we document, data breaches, ransomware incidents, and disruptions to digital infrastructure now occur at an alarming frequency, with thousands of major cases recorded each year in the United States alone. These events inflict direct financial losses on firms[1] and ripple through the economy by disrupting supply chains, eroding customer trust, and weighing on productivity and investment. Because cyberattacks impose substantial costs, they affect firms' incentives to allocate resources, and can alter decisions about entry and exit. Accordingly, policymakers are taking notice[2]. However, despite its growing importance[3], the macroeconomic consequences of rising cyber risk remain largely unexplored.

While existing research has made important progress in documenting firm-level consequences, such as declines in market valuation and investment[4], and in modelling cybersecurity decisions in static, partial-equilibrium settings[5], these approaches fall short of providing a framework in which cyber risk can be evaluated as a systemic phenomenon and policy trade-offs can be analyzed quantitatively. In particular, these approaches do not account for three key factors: the process by which firms and attackers meet, strategic considerations on the part of attackers and firms, and general equilibrium effects.

In this paper, we propose such a framework and quantify it using novel micro data. Our primary contribution is to develop a micro-founded, dynamic general equilibrium model of cyber risk in which interactions between firms and attackers are governed by a search-and-matching process. Search frictions determine how attackers and firms meet, and congestion limits attackers' ability to concentrate exclusively on the most attractive targets. Whereas the literature takes the meeting process between firms and attackers as exogenous, this paper models it explicitly, making it respond endogenously to economic variables, including the firm distribution and cyber technology. Firms choose cyberse-

[1] The Ponemon Institute and IBM (2025) estimate the average cost of a breach was $4.4 million in 2024.

[2] The financial sector is particularly prone to cyberattacks and also highly vulnerable given the interconnected nature of financial activities, leading to concerns from central banks and financial regulators. For instance, the Federal Reserve annually publishes a Cybersecurity and Financial System Resilience Report to Congress on actions taken to strengthen the security of the financial sector (Board of Governors of the Federal Reserve System 2025). Broader concerns have been raised around the security of critical infrastructure projects and national security.

[3] While we do not explicitly analyse the consequences of Artificial Intelligence (AI) adoption in this paper, we expect AI to amplify both the frequency and severity of cyberattacks by lowering entry barriers for attackers in developing exploits, and also to aid defenders in shoring up cybersecurity levels by identifying vulnerabilities earlier. A full analysis of the impact of AI on cyber risk is beyond the scope of this project, and is left for future work.

[4] See, for instance, Kamiya et al. (e.g., 2021); Akyildirim et al. (e.g., 2024).

[5] See, for instance, Moore, Clayton and Anderson (2009); Ramírez (2025); Ebel and Mitra (2024).

curity investment in anticipation of attacks, while attackers choose which types of firms to target and how intensely to strike them. Firms' cybersecurity investments therefore depend on their expected exposure to attacks, while attackers direct their efforts toward firms whose size and defenses make them most attractive. Thus, realized cyber risk reflects a fixed point in the actions taken by firms to defend themselves and the actions of attackers. These strategic interactions shape firms' entry and exit decisions and their exposure to attack, altering the equilibrium firm size distribution and amplifying the aggregate consequences of cyber risk through general equilibrium.

Model. The model integrates two building blocks of modern macroeconomics: a search-and-matching framework à la Diamond (1982), and Mortensen and Pissarides (1994) and an endogenous firm entry–exit margin à la Hopenhayn (1992). Markets are segmented by firm productivity, attackers direct their search toward particular firm types, and free entry of attackers determines the rate at which attackers and firms meet. At the same time, firms decide whether to incur a fixed cost to enter the market. Those that enter hire labor, produce output, and invest in cybersecurity to defend against potential attacks. Production is subject to fixed costs, which occasionally lead firms to exit. When attackers strike they choose the intensity of their attacks against firms of a given size.

A matching technology determines the probability that an attacker meets a firm, which therefore depends on the "tightness" of the cyber market, the ratio of the number of vacant attackers to potential targets.[6] Upon matching, attackers decide the intensity of their attack subject to a capacity constraint, the highest possible attack intensity, which we call *attack capacity*. Attacks succeed with a probability increasing in attack intensity and decreasing in the level of cybersecurity chosen by the targeted firm. Successful attacks reduce firm output over their duration and allow the attacker to extract rents increasing in the size of the targeted firm. Finally, we introduce an externality from cyberattacks by having aggregate productivity decrease in the share of attacked firms.

Empirics and quantification. The model is calibrated to match several aspects of the U.S. economy, including standard firm-related moments and cyber-related moments taken from several datasets or the literature. A key empirical contribution of the paper is to document how cybersecurity investment varies with firm size. We quantify it using LinkedIn employment profiles provided by Revelio Labs. Using worker job titles linked to the firm

[6]The role of attackers is analogous to that of vacancy-posting firms in search and matching models of the labor market, while firms play the role of unemployed workers meeting attackers at random. However, while it is typically assumed that unemployed workers desire matches with posted vacancies, firms in our model would rather not be contacted by cyberattackers.

they are employed at, we show that the share of workers employed in cybersecurity occupations rises quickly with firm size, a fact that is robust across industry and time. Furthermore, smaller firms do not compensate their lower in-house cyber investments by outsourcing their cyber services more than larger firms. We use this empirical gradient to discipline cybersecurity investment in the model.

This empirical pattern is central to the model's implications for attack risk. In the model, consistent with the available evidence, the incidence of attacks follows an inverse-U shape in firm size. The smallest firms are unattractive targets because their payoff to attackers is limited, while the largest firms invest heavily in cybersecurity and are therefore difficult to breach. Mid-sized firms, in contrast, represent both a meaningful prize and a tractable target, making them disproportionately vulnerable. We provide empirical evidence for this claim using rich microdata on individual breaches[7].

In addition, our model emphasizes the importance of general equilibrium effects operating via entry and exit, and the labor market, that are absent from partial-equilibrium analyses. In particular, cyber risk shifts the firm size distribution toward smaller firms, lowering aggregate productivity. When the model is calibrated to U.S. data, the introduction of cyber risk leads to a decline in firm entry of 3.6 percent, in aggregate productivity of 0.6 percent, and in output and consumption of 1.8 and 1.7 percent, respectively, relative to an economy without cyber risk.

Rising cyber risk. In counterfactual exercises, we consider two scenarios likely to characterize the cyber landscape in the future. To capture the advent of AI empowering attackers, we consider a scenario in which attacks are harder to counter; to capture the rising digitalization of the economy we consider a scenario in which the match probability is higher. We model the former by an increase in the attack capacity of the typical attacker, and the latter by an increase in the efficiency of the matching technology determining firm-attacker contact rates. We use these experiments to show how evaluating the consequences of a rise in cyber risk requires understanding its source, the endogenous responses of economic actors, and the general equilibrium consequences. What we find is that cyber risk is most damaging when it expands the set of exposed firms.

[7]We note that empirical validation of this claim is severely complicated by the fact that cyberattacks remain among the most under-reported category of crimes, and that mandated disclosures are heavily skewed toward larger and public firms. Taking our model seriously would suggest that the under-reporting of cyberattacks is highest among medium-sized businesses, and that the collection of survey data on this category of firms is of critical importance. The Ponemon Institute's 2016 State of Cybersecurity in Small and Medium-Sized Business surveyed 598 businesses with between 100 and 1000 employees, finding that 55% of respondents claimed a cyber attack in the previous year, a number much higher than suggested by datasets tracking cyberattacks (Ponemon Institute 2016).

Policy implications. Beyond positive analysis, our model allows us to consider normative questions. Cybersecurity investments are costly, but their benefits extend beyond the individual firm, since reduced vulnerability improves outcomes for trading partners and customers and lowers aggregate attack probabilities. This externality provides a rationale for government intervention. We explore three policies governments may adopt to mitigate cyber risk: firm subsidies for cybersecurity investments, minimum cybersecurity regulatory requirements, and ex-post bailouts to attacked firms. We find that appropriate subsidies and regulation can raise aggregate output. In particular, modest and homogeneous regulatory requirements are the most robust policy across a range of cyber-risk environments, because they raise cybersecurity investment among small and mid-sized firms, which are relatively underprotected, while leaving firms that already invest heavily largely unconstrained. In our benchmark economy, the output-maximizing minimum requires firms to devote just 0.2% of labor to cybersecurity, yet raises the aggregate cybersecurity labor share by 65 percent. Our results highlight that policy evaluation in the cyber domain must account for direct, indirect, and equilibrium effects. The costs of cybersecurity investment, the cyber externality, and firm entry and exit together shape aggregate outcomes.

Related literature. Our paper contributes to a nascent but fast-growing literature on cybersecurity. On the empirical side, several studies have measured the costs of breaches, including reductions in firm revenues, investment, and market valuations (Romanosky 2016; Anderson et al. 2013). A related strand documents the distribution and dynamics of breaches across industries and firm types (Edwards, Hofmeyr and Forrest 2016; Seh et al. 2020; Barati and Yankson 2022; Carfora and Orlando 2022), with recent work emphasizing the disproportionate vulnerability of medium-sized firms (Baksy, Caratelli and Olson 2025). Florackis et al. (2023) and Jamilov, Rey and Tahoun (2023) study how cyber risk exposure varies across firms, how cyber risk affects asset prices and can generate substantial spillover effects via financial markets.

Another strand emphasizes the financial stability and macroprudential implications of cyber risk. Duffie and Younger (2019) point to the real economic effects cyberattacks can have through shared digital infrastructure and payment systems. Eisenbach, Kovner and Lee (2022) highlight how cyber shocks differ from traditional operational risks, arguing that their systemic nature warrants dedicated modeling frameworks and stresses the potential amplification channels through payment and settlement systems when a cyber shock hits the financial system. Using a natural experiment, Kotidis and Schreft (2025) provide causal evidence that cyber incidents propagate through financial networks, rais-

ing funding costs and weakening balance sheets of firms not directly attacked, highlighting an important channel through which cyber shocks generate systemic risk. Koo et al. (2022) develop a framework to integrate cyber events into macro-financial risk monitoring, demonstrating how breach data can inform stress testing exercises. Anand, Duley and Gai (2026) emphasise the importance of accounting for spillovers via shared software platforms.

While both of these literatures provide compelling evidence on the firm-level consequences of cyberattacks and their propagation, they do not offer a framework to quantify their aggregate effects or to evaluate policy in equilibrium. A contribution of this paper is to provide a micro-founded equilibrium framework that links micro-level cyber interactions to aggregate outcomes. It does so by embedding a game-theoretic treatment of cyberattacks and a search-and-matching framework modelling the way attackers and firms meet into a dynamic general equilibrium model of firm entry and exit.

On the theory side, there is a long precedent for modeling attack and defense as strategic resource-allocation problems. Classic max–min formulations of weapons-allocation problems study how defensive resources should be allocated across targets in anticipation of an optimizing adversary (Danskin 1967). More recently, a set of papers model cyberattacks as strategic interactions between attackers and firms (Moore, Clayton and Anderson 2009; Ramírez 2025), though typically in static and partial-equilibrium environments. This limits their ability to capture how these interactions shape entry, firm size, and aggregate outcomes. Our framework extends this approach by embedding these interactions in a dynamic general equilibrium with endogenous firm entry and exit, allowing both firm behavior and attacker activity to adjust endogenously.

Existing models (e.g. Ebel and Mitra 2024) of strategic cyber interaction typically study the allocation of attack and defense resources conditional on an attacker confronting a potential target. Our framework introduces an additional layer: attackers must search for and match with firms before an attack can take place. This allows the model to endogenize not only attack intensity conditional on a target, but also attacker entry, target choice, contact rates, and congestion across targets.

The model allows us to evaluate policy interventions aimed at mitigating the macroeconomic consequences of rising cyber risk. Our results indicate that private incentives for cybersecurity investment can diverge from social incentives. This echoes the findings of Erol and Lee (2024), who show that competition among financial market infrastructures can lead to underinvestment in technological resiliency and system-wide "technological drag." We extend this insight to a general equilibrium, macroeconomic model, showing how cyber risk shapes firm entry, the size distribution, and aggregate productivity.

2 Cyber Risk: What You Need to Know

This section documents five empirical facts that motivate the model. First, cyber risk has increased substantially over the last two decades, both in the number of reported incidents and in their severity. Second, this rise has not been confined to the largest firms: several measures suggest that cyber incidence has increased especially among medium-sized firms. Third, we show that the share of cybersecurity employment scales rapidly with firm size. This fact is central to the mechanism in the model, because larger firms are more attractive targets but also invest more intensively in protection. Fourth, we show that most cyberattacks are motivated by financial gain, supporting the modeling of attackers as economically motivated agents. Fifth, we review evidence that successful cyberattacks can be highly disruptive, with costs that extend beyond the immediate breach.

Before proceeding, we emphasize an important measurement caveat. Cybercrime is selectively reported and difficult to quantify, so observed breach data likely understate true attack incidence. This problem is especially severe for smaller and private firms (Romanosky, Telang and Acquisti 2011). This limitation motivates our use of multiple data sources in this section and, in the sections that follow, our structural approach.

2.1 Cyber risk has increased substantially over the last 20 years

Cybersecurity threats have increased substantially since 2005 in the United States. To establish this we use the *Data Breach Chronology*, compiled by the Privacy Rights Clearinghouse (PRC). These data provide a broad but incomplete record of data breaches in the United States[8]. The dataset records information from a wide range of sources, including state attorneys-general, regulatory filings, media reports, and company press releases, and records details on each event. These include the identity and type of the affected organization, the date the breach was reported, the estimated number of individuals impacted, the type of breach, the sector of the affected entity, and the duration of the incident. The data contain 74,783 breaches; we focus on the 27,385 attacks which occurred between 2005 and 2024 inclusive, in the United States, and affected private business organisations.

Figure 1 shows that the total number of reported attacks has increased by several orders of magnitude over the past 20 years, rising from 17 attacks in 2006 to almost 3,000 in 2023. The rapid increase in the number of attacks is paralleled by a rise in their severity,

[8]The Privacy Rights Clearinghouse Data Breach Chronology has been widely employed in research on cybersecurity. Examples include and risk modeling (Carfora and Orlando 2022) and breach impact analysis (Rosati 2021; Barati and Yankson 2022; Edwards, Hofmeyr and Forrest 2016).

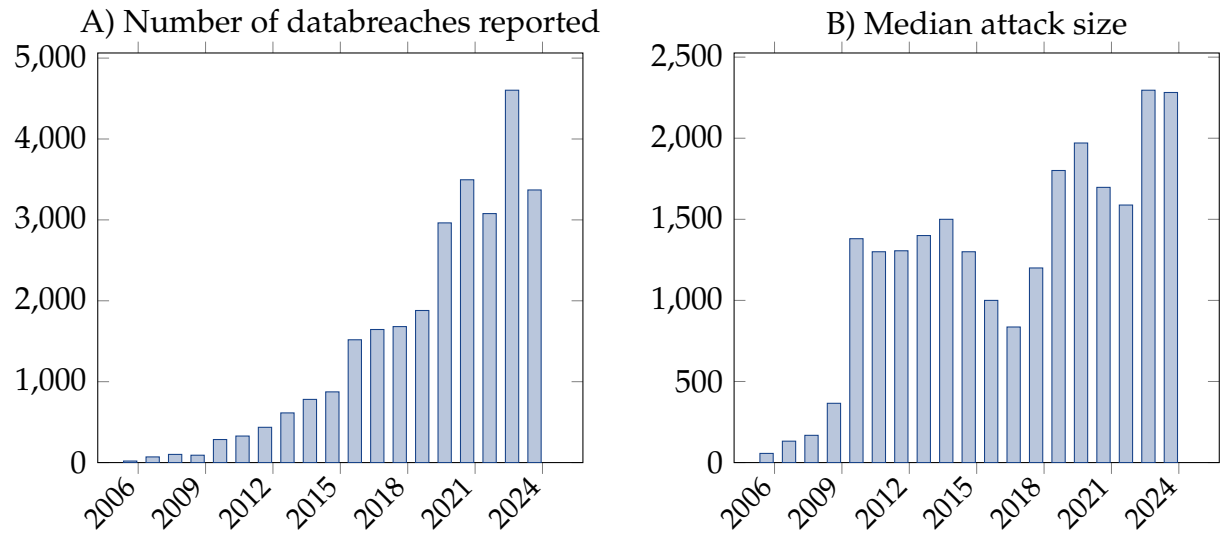


Figure 1: Number of attacks reported in the US (panel A) and the median number of records breached across breaches in a given year (panel B). All data from the Privacy Rights Clearinghouse.

measured as the median number of records compromised.

2.2 Incidence has risen for medium-sized firms

The rise of large, data-intensive corporations has increased the concentration of valuable digital information within firms (Baley and Veldkamp 2025; Begenau, Farboodi and Veldkamp 2018*a*). This raises the possibility that the increase in cyberattacks has been concentrated among the largest firms. We cannot test this directly in our database because it identifies attack targets by name but does not contain information on target size.[9] Nevertheless, we provide two pieces of evidence indicating that the rise in attacks increasingly affects mid-sized firms.

First, while the DBC does not directly provide a measure of firm size, it does provide an estimate of the number of records breached in an incident; under the assumption that this number scales with firm revenue[10] and hence with firm size, the distribution of attacks across the number of breached records is informative about the distribution of

[9]We have explored matching the dataset with Compustat to obtain firm sizes using fuzzy matching algorithms on firm names. This exercise has not proved fruitful for two reasons. First, the match rate between Compustat and the DBC is persistently low, since the former contains only public firms whereas the latter contains many small, private firms. Second, employment information in Compustat is only available annually and includes employees from non-US locations, making it difficult to measure domestic firm size.

[10]The assumption that the size of a firm's database is related positively to the firm's size features in recent theoretical and quantitative work on the role of data in determining the firm size distribution (Begenau, Farboodi and Veldkamp 2018*b*; Farboodi et al. 2019).

attacks by firm size. Figure 2 depicts the distribution of attacks across bins of the number of records breached, year-on-year. The figure makes it clear that the rising mean number of breached records is not driven by headline-grabbing breaches compromising over a million records, which are likely to affect large corporations. Instead, the rise is driven by a stark increase in attacks compromising an above-average 1,000 to 10,000 records, the sizes of databases associated with mid-sized enterprises: such attacks have grown from comprising about 6.3% of all attacks to nearly 34% of them. Attacks involving the largest numbers of record breaches (over a million) account for only about 2% of all breaches, a share that has remained stable over our sample period.

Given that reporting standards in some cases require larger firms to report cyberattacks in greater detail, and that the coverage of our data is likelier to omit small and medium sized firms, this is likely a stark underestimate of the actual share of attacks that small and medium enterprises have faced. For instance, the Ponemon Institute's Review of Small and Medium-Sized Business cybersecurity in 2016 (Ponemon Institute 2016) found that 55% of all medium-sized firms (between 100-1000 employees) experienced a cyberattack in the year prior to the survey.

We provide additional evidence on the vulnerability of small and medium firms using data from the VERIS[11] Community Database, constructed by the Verizon RISK team using information crowdsourced from partners. The VERIS dataset has the advantage of including information on firm size categories and of capturing cyber attacks other than data breaches, including ransomware attacks. Figure 3 shows that for private-sector organisations primarily operating in the US, between 2010 and 2022[12], firms with employment between 101 and 1000 increased their share of reported incidents over two-fold from around 16% to nearly 36%.

2.3 Investments in cybersecurity scale rapidly with firm size

Our main empirical contribution is to document how cybersecurity investment scales with firm size. Estimates of the amount spent on cybersecurity investments are rare, and generally obtained by proprietary surveys which under-represent small firms[13]. There-

[11]VERIS stands for the Vocabulary for Event Recording and Incident Sharing, and represents a unified framework within which organisations can anonymously share information about security incidents. The data is publicly available on GitHub and is used by Verizon to produce its annual Data Breach Investigations Reports, published since 2008.

[12]Prior to 2010, the number of reported incidents in the VCDB is too low for us to draw meaningful conclusions.

[13]For instance, Moody's Ratings (2023) estimates that corporations spent a median share of 8% of their IT budgets on cybersecurity, with a range of 5-10% across sectors. However, their sampling frame is firms whose debt issuances are rated by Moody's, which is likely to overrepresent large firms.

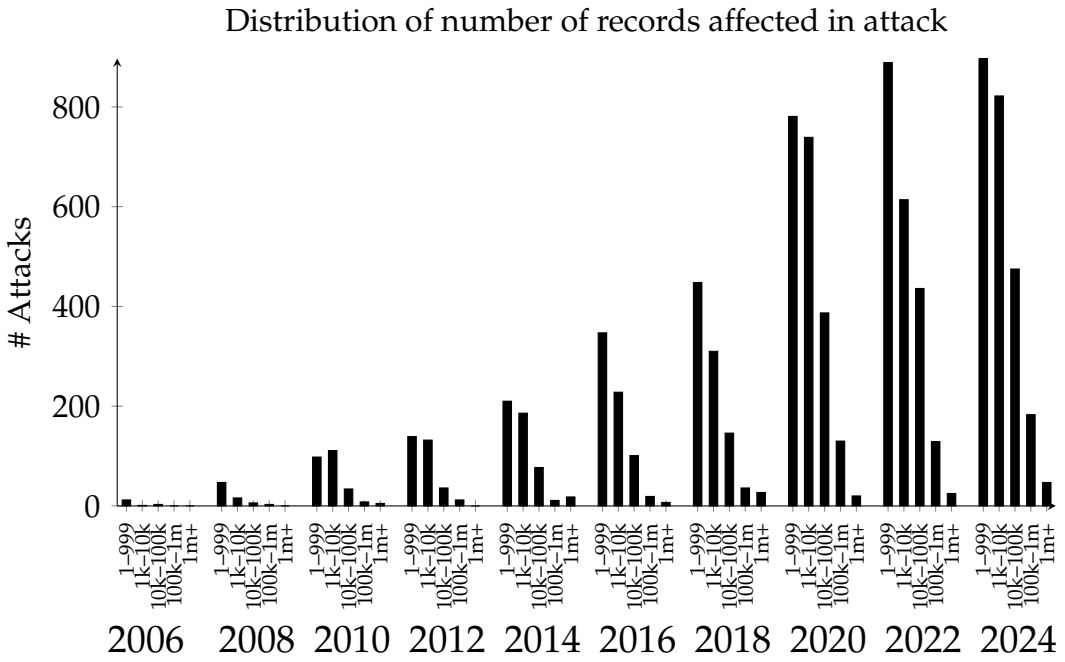


Figure 2: Distribution of attacks across record loss classes. All data from the Privacy Rights Clearinghouse.

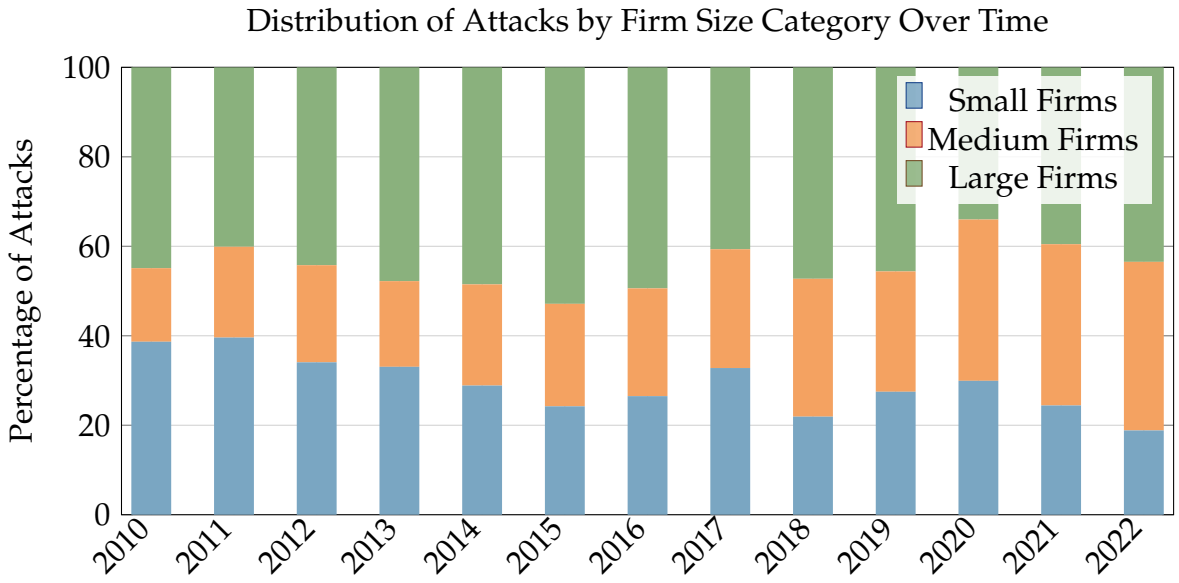


Figure 3: Distribution of attacks across firm size classes. Small firms: 1-99 employees, medium firms: 100-999 employees, large firms: 1000+ employees. All data from the VERIS database maintained by Verizon's RISK Team.

fore, to study how cybersecurity investments scale with firm size, we rely on data on the employment of cybersecurity professionals across firms. Our data comes from LinkedIn profiles compiled by Revelio labs, a provider of workforce intelligence. Revelio Labs' data covers nearly 90 million worker profiles a year over the 25-year period we study, including over 290,000 cybersecurity-related job spells, across nearly 8 million unique firms. Our sample is constructed using worker-firm job spells with a valid occupation, starting date and ending date between January 2000 and November 2024, when our data was obtained. We exclude all observations associated with firms that do not contain a valid Revelio identifier, but do not otherwise restrict our sample. We rely on Revelio's `k1500` occupation classification and identify cybersecurity related job spells by searching for titles containing the strings combining "{IT, information, cyber, network}" with "security", and "security" with "{analyst, architect and engineer}". Our results are robust to a wide range of alternative specifications, including one where we rely only on occupations mapped to the O*NET code 15.1212, Information security analysts.

Figure 4 shows a clear pattern: larger firms devote a higher share of employment to cybersecurity, with a pronounced rise among firms with more than 10,000 employees since the late 2010s. Table 1 shows that this correlation remains positive even when we consider variation within industries and firms over time, with a remarkably stable semi-elasticity - a firm that is twice as large has a cybersecurity share of its workforce that is almost a percentage point larger.

Variable	Share of cybersecurity-related employment		
log(Total Workers)	0.0128***	0.0092***	0.0167***
	(0.0002)	(0.0002)	(0.0004)
Year FE	No	Yes	Yes
Industry FE	No	Yes	No
Firm FE	No	No	Yes

Table 1: Estimates of the semi-elasticity of the share of cybersecurity-related employees to firm size. All regressions are based on the set of firms in Revelio Labs data reporting at least one US employee with a valid industry code and firm identifier. Firm size is the number of employees of a firm recorded in the Revelio Labs dataset.

One concern with this measure of cybersecurity investments is that it misses alternative sources of investment not captured by direct employment, and that these may systematically differ across the firm-size distribution. In particular, small firms may choose to *outsource* their cybersecurity to dedicated providers rather than develop in-house capabilities. We provide evidence that this is not the case using rich microdata from the

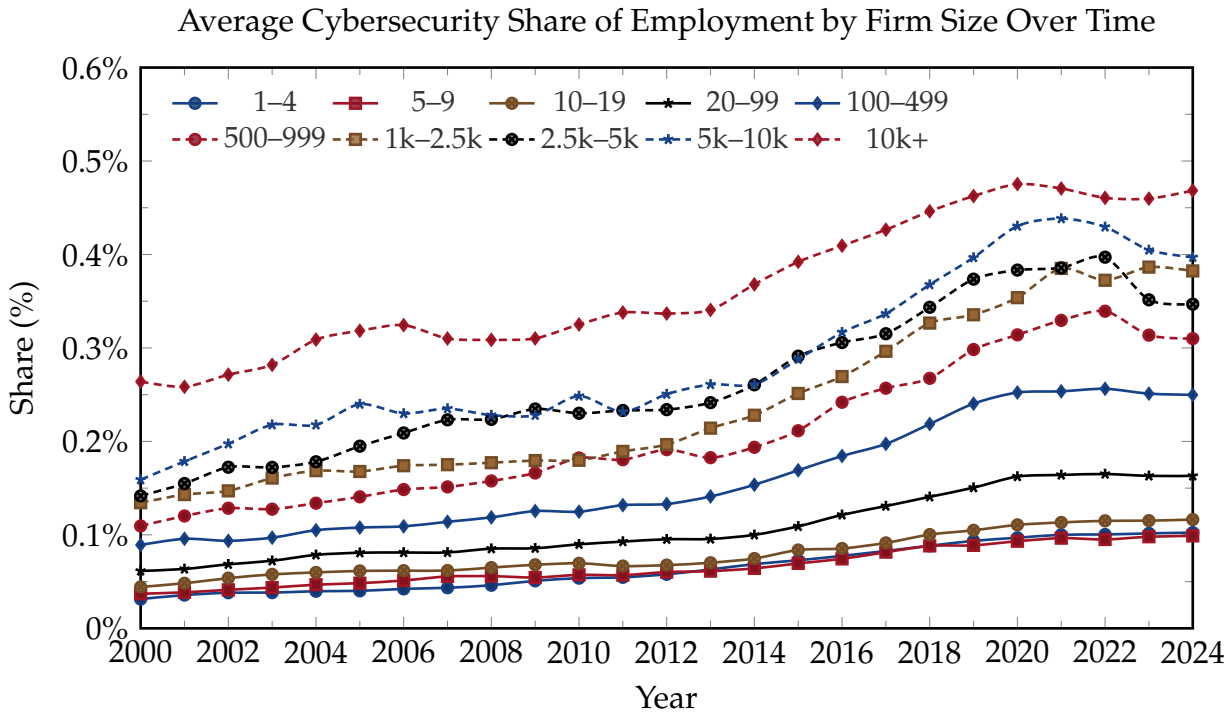


Figure 4: Share of cybersecurity-related employment across firm-size categories. Source: LinkedIn/Revelio Labs and authors' calculations.

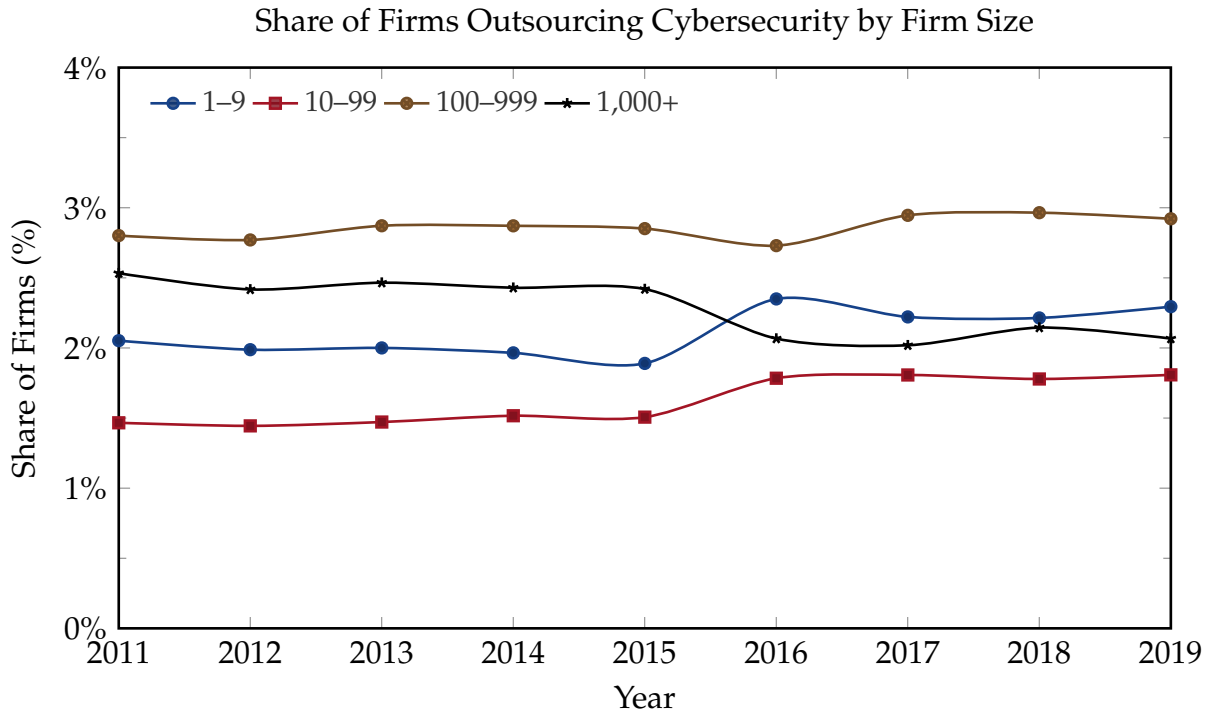


Figure 5: Share of establishments by employment size bins reporting outsourcing of firewall or intruder detection/protection system maintenance, 2011-2019. All data from the Computer Intelligence Technology Database.

Computer Intelligence Technology Database (CiTDB) produced by Aberdeen/Zipf Spiceworks Davis[14]. Based on survey data collected directly from nearly 226,000 unique establishments for IT technology marketing purposes[15], the database includes, between 2011 and 2019, information on whether a given establishment outsources firewall maintenance or the maintenance of its intrusion detection and protection systems (IDS/IPS). Figure 5 shows that the share of firms reporting that they rely on outsourcing of firewall or IDS/IPS maintenance does not vary much across firm size categories, and that this pattern has persisted over time. If anything, the plot figure shows, it is mid-to-large firms that tend to outsource the most.[16] Leveraging the full richness of the microdata, we see that larger firms are slightly more likely to outsource cybersecurity services and this is not driven by industry fixed effects.[17]

Variable	Any cybersecurity maintenance outsourced	
log(No. of Workers)	0.0038***	0.0035***
	(0.0002)	(0.0002)
Year FE	No	Yes
Industry FE	No	Yes

Table 2: Linear probability regressions using 1,517,810 firm-year observations from 2011–2019. The outcome equals one when outsourced firewall or intrusion-detection maintenance is recorded. Standard errors in parentheses are clustered by firm. *** denotes $p < 0.001$.

2.4 Cyberattacks are motivated by financial gain

Because our model emphasizes the financial incentives of firms and attackers, it is important to establish that cyberattacks are not driven only by geopolitical conflict, espionage, or other non-economic motives. Economically damaging cyberattacks may have a

[14]We use a historical archive of the dataset originally collected for use in the article Bloom, Draca and Van Reenen (2016), and sincerely thank Nick Bloom for permission to use the data.

[15]Historical versions of the data have been used in the economics literature to track the build-out of information and communication technologies during the boom of the 2000s (Bresnahan, Brynjolfsson and Hitt 2002; Forman, Goldfarb and Greenstein 2012; Hershbein and Kahn 2018)

[16]Our estimates of cybersecurity outsourcing are much lower than those reported by industry participants. We note, however, that (1) our data is based on establishment-level information, while reported numbers are firm-level, and (2) our data is broadly representative of the population of establishments, while survey evidence is skewed toward digital-native sectors and larger firms. Nonetheless, estimates strongly support the conclusion that small establishments are *less* likely to outsource cybersecurity functions than larger ones; see here, for instance: https://www.ipsos.com/sites/default/files/2017-04/sri-cybersecurity-breaches-survey-2017.pdf

[17]There is no variation across our sample in outsourcing by firm over time.

wide range of objectives, including non-financial ones. However, the available evidence suggests that financial gain is the predominant motive behind recorded cyberattacks.

The Privacy Rights Clearinghouse data do not contain a direct measure of attacker motives, making it difficult for us to distinguish between attacks primarily motivated by financial gain and attacks driven by non-financial objectives.[18] We therefore rely on a dataset of cyberattacks maintained by the Center for International and Security Studies at Maryland (CISSM). Table 3 shows that, between 2014 and 2023, roughly 89% of all cyberattacks in the dataset were classified as financially motivated.

Year	Financial	Espionage	Personal	Protest	Political	Sabotage
2014	103	14	13	19	9	1
2015	95	9	7	28	9	1
2016	146	5	8	33	5	10
2017	216	16	11	8	14	12
2018	293	29	5	8	12	1
2019	333	11	3	3	7	5
2020	486	26	5	6	20	3
2021	529	12	1	15	9	1
2022	747	6	0	32	6	1
2023	1052	8	0	23	8	6

Table 3: Cyberattack motives in the CISSM dataset. We pool espionage targeted against firms and governments into one category for parsimony.

2.5 Successful cyberattacks can be extremely disruptive

Estimates of the indirect losses from cyberattacks vary widely due to severe data challenges and issues around defining the exact nature of these losses[19] and yet it is generally accepted that the indirect losses dominate direct losses. Cybersecurity and Infrastructure Security Agency (2020) describe in detail the challenges associated with measuring the costs of a cyberattack and in reconciling the wide range of estimates available. Cobos and Cakir (2024) provide a rich review of the literature on estimating the costs of an

[18]The Office for Financial Research's Annual Report to Congress (Office of Financial Research 2022) lists multiple possible alternative motivations that attackers may have, including revenge, reputation, geopolitics, and international security.

[19]Kamiya et al. (2021) show that the costs associated with investigation, remediation, legal and regulatory penalties associated with 75 major first-time cyberattacks were dominated by the loss in shareholder value by a factor of nearly 100. It is, however, unclear whether the entire loss in shareholder value, $104 billion, estimated by the authors should be attributed to the attack alone. For instance, susceptibility to a cyberattack may provide investors with adverse information about the resilience of a firm to alternative forms of risk, hence depressing its market value.

attack, discussing evidence on losses driven by negative stock market reactions and reputational effects, the impacts of supply chain disruptions and spillover effects, and the delayed announcement of breaches. Their findings suggest that the economic costs of cyber incidents are substantial, but highly uncertain: estimates of aggregate global cyber costs range from hundreds of billions of dollars annually to several trillion dollars. More credible firm-level evidence points to economically meaningful losses: data breaches cost firms about $4 to $5 million on average, while event-study estimates typically find stock-market losses of roughly 1 to 2 percent around breach announcements, with larger losses for some types of incidents.

Relying on data from the NotPetya breach in 2017, Crosignani, Macchiavelli and Silva (2023) show that supply chain propagation of the effects of a major cyberattack can result in a four-fold amplification of the initial shock, which represents an empirical target for parameters governing spillover effects in our model below.

3 Macroeconomic Model of Cyber Attacks

There exists no established general equilibrium macroeconomic model to analyze cyber attacks. In this section we build one.

Model overview. In the model firms with heterogeneous productivities are attacked by cyber agents and can choose to defend themselves through cybersecurity investments. The model nests a search and matching framework within a model of firm entry and exit à la Hopenhayn (1992). Time is discrete, and indexed by $t = 0, 1, \ldots$. There is one final good in the economy which is used solely for consumption and is the numeraire. There are three sets of agents, households, firms and cyberattackers. Households supply labor to firms, which use labor to produce the final good using a technology featuring strict diminishing returns. All markets are competitive.

Firms are subject to occasional cyberattacks, which disrupt production and have spillover effects on aggregate productivity. To avoid the effects of attacks, firms make upfront investments in cybersecurity. In each period, attackers in our economy pay a cost to search for firms of a given productivity level. Our setup is therefore analogous to a directed search model in which "markets" are segmented by productivity. Within each market, the probability of a firm and a cyberattacker meeting depends on the "tightness" of this market, the number of firms with that productivity level divided by the number of attackers targeting such firms.

When a firm and attacker meet, the latter decides on the quantity of resources it will

devote to attacking the firm, subject to a limit. The outcome of a firm-attacker meeting depends on the level of cybersecurity and the chosen attack intensity; the higher the intensity of the attack relative to the cybersecurity level, the more likely an attack is to be *successful*. Successful attacks result in a loss of revenue for the firm and a positive share of revenue captured by the attacker.

In equilibrium, firms anticipate the likelihood of an attack and choose their ex-ante cybersecurity investments accordingly, trading off a convex cost of such investments against the value of a reduced likelihood of being attacked. Attackers anticipate the level of investment the typical firm of a given productivity level is likely to make, targeting their attacks against only those firm types whose bounty is sufficiently attractive and whose investment level is not excessively high. Fixed costs induce relatively low productivity firms to exit, and higher likelihoods of being attacked reduce the incentives of firms to enter, together shaping the equilibrium firm size distributions and hence determining aggregate productivity and output. Throughout, we focus on steady states in which macroeconomic aggregates are constant and firm size distributions are stable.

3.1 Firms

There is a mass of firms that produce a homogeneous good, which is the numeraire. Firms use labor as the only input into production. There is one unit of inelastically supplied labor whose price is w in the whole economy. A firm with productivity φ produces according to the production function $y(Z, \varphi, n) = Z\varphi n^{\alpha}$ where n is the firm's labor demand and $\alpha \in (0, 1)$ is the degree of returns to scale. Z is an aggregate productivity term all firms in the economy are subject to – we will return to Z later in detail. In addition to labor costs, firms must pay a per-period fixed cost f to operate. Firms discount the future with factor β and are hit with an exogenous exit shock with probability σ. Finally, firms' productivity follows a first-order Markov process with transition probability $\Pi(\varphi'|\varphi)$, which is a discretization of the following AR(1) process:

$$\log(\varphi') \quad = \rho_z \log(\varphi) + \epsilon \tag{1}$$

where ρ_z is the persistence of the process and $\epsilon \sim \mathcal{N}(0, \sigma_z)$ is the innovation hitting productivity each period.

Safe firms. A *safe* firm is one that is not currently under a cyber attack. Safe firms choose the amount of labor to employ and the size of the cyber security investment, $x \geq 0$, which we assume is denominated in units of labour. We assume that cyber hiring x leads to

cybersecurity level $C(x)$, where $C' > 0, C'' < 0$. The value of a safe firm with productivity φ is

$$\begin{aligned} V^s(Z,\varphi) &= \max_{n,x\geq 0} Z\varphi n^\alpha - wn - wx - f \\ &\quad +\beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[q(Z',\varphi')V^m\left(Z',\varphi',x\right) + \left(1-q(Z',\varphi')\right)V^s(Z',\varphi')\right] \quad (2) \end{aligned}$$

What awaits the firm in the next period, if it survives, is one of two states: with probability $(1-q(Z',\varphi'))$ the firm remains safe, and with probability $q(Z',\varphi')$ the firm meets a cyber attacker and derives value $V^m(Z',\varphi',x)$. We characterize V^m below.

Since there are no adjustment costs or search frictions in the labor market, the choice of labor is effectively a static one that solves the profit maximization problem

$$\pi(Z,\varphi;w) \equiv \max_{n\geq 0} Z\varphi n^\alpha - wn$$

which yields the labor demand curve for a firm, $n(Z,\varphi;w) = \left(\frac{w}{\alpha Z\varphi}\right)^{\frac{1}{\alpha-1}}$. The maximized profits for a firm are given by

$$\pi(Z,\varphi;w) = Z\varphi^{1-\alpha}\left(1-\alpha\right)\left(\frac{w}{\alpha}\right)^{\frac{\alpha}{\alpha-1}}$$

The firm's choice of cybersecurity investment x is given by the FOC equating the marginal benefit and marginal cost of cyber security,

$$w = \beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[q(Z',\varphi')\frac{\partial V^m(Z',\varphi',x)}{\partial x}\right] \quad (3)$$

We can interpret the left hand side as the marginal cost of investing in cybersecurity, namely the wage paid to the labor invested in cyber security. The right-hand side is the benefit of this investment discounted by the firm's effective discount factor taking survival into account. We will describe this benefit next. We assume that cybersecurity investments must be made on an ongoing basis, and that the level of cybersecurity at date t is entirely determined by investments made at $t-1$. Assuming instead that cybersecurity levels were persistent would add a state variable to the model without substantially impacting the economics of our results. Further, given the continual emergence of new threats faced by firms, we expect that the level of cyber protection a firm has must be continuously refreshed in order to remain effective, which lends credence to our modelling approach.

Firms under attack. Meeting a cyber attacker does not imply a cyber attack will be successful. When attacker and firm meet, the value to the firm is

$$V^m(Z,\varphi,x) \;=\; \Lambda\left(a^*(x),x\right)V^a(Z,\varphi)+\left[1-\Lambda\left(a^*(x),x\right)\right]V^s(Z,\varphi) \tag{4}$$

With probability $\Lambda\left(a^*(x),x\right)$, where $a^*(x)$ is the optimal choice of attack intensity by the attacker, the firm is successfully attacked and gets value $V^a(Z,\varphi)$; otherwise the firm is safe. Note that the cyber security investment is *predetermined*, capturing the idea that cyber security investments must be ongoing and act as insurance that protects against attacks when they occur. We will describe the determination of Λ below.

When a firm is successfully attacked, it loses a share ℓ of its output but must continue to pay the fixed cost of production and its labor bill. To capture the fact that cyber attacks have persistent consequences, the attacked firm escapes the attack with probability δ. If the firm chooses to, it can exit altogether. The value of a firm under attack is

$$\begin{aligned} V^a(Z,\varphi) = \max\big\{0,\ \max_n (1-\ell)\cdot Z\varphi n^\alpha - wn - f \\ + \beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[(1-\delta)V^a(Z',\varphi') + \delta V^s(Z',\varphi')\right]\big\} \end{aligned} \tag{5}$$

We note that the profit maximization problem pinning down the choice of labor by the firm is isomorphic to that of a safe firm with a reduced productivity level of $(1-\ell)Z\varphi$, facilitating the computation of the labor demand, $n^a(Z,\varphi;w)$, and profit functions, $\pi^a(Z,\varphi;w)$. By the Envelope theorem, we have from equation (3) that

$$w = \beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[q(Z',\varphi')\left[\frac{-\partial\Lambda\left(a^*(x),x\right)}{\partial x}\right]\left(V^s(Z',\varphi')-V^a(Z',\varphi')\right)\right] \tag{6}$$

This first order condition is the central equation describing the trade-offs faced by firms in deciding on the level of cybersecurity they will adopt. On the left side, we have the marginal cost of an additional cyber worker hired. On the right hand, we have the marginal benefit, consisting of the discounted present value net of exogenous exit of the added security this worker brings. The value of an added cyber worker is the reduction in the probability of a successful attack conditional on matching, times the difference in firm value depending on whether the firm is safe or under attack. Forces that raise the cost of labour make cybersecurity more expensive, reducing expenditure on security *ceteris paribus*. Forces that raise the matching probability, or the conditional probability of a successful attack conditional on a match, will tend to raise cyber investments in general; however, forces that reduce the value of being a safe firm relative to that of being a firm

under attack will also reduce investments. This is an important mechanism in explaining our results below.

Firm entry and exit. There is a large pool of prospective entrant firms. To enter the market, firms must first pay an entry cost f_e and draw a productivity φ from a Pareto distribution with scale parameter ϖ whose (discretized) probability mass function is $g(\varphi)$ with (discretized) cumulative distribution function is $G(\varphi)$ and whose productivity states are $\{\varphi_1, \ldots \varphi_P\}$.

At this stage, firm φ can choose whether to become *operational*, or exit the market. The value of an entrant with productivity φ is

$$V^e(Z, \varphi) = \max\{0, V^s(Z, \varphi)\} \tag{7}$$

What emerges from this is that there exists a set $\varphi = \{\varphi : V^e(Z, \varphi) \geq 0\}$ that consists of the profitable productivities whereas firms with $\varphi \notin \varphi$ will immediately exit.[20]

Because there is an unbounded set of prospective entrants, the free entry condition for firms in this economy is

$$\sum_{p \in \varphi} V^e(Z, \varphi)\, g(\varphi) - f_e = 0 \tag{8}$$

Because productivity is stochastic, safe firms may *endogenously* choose to exit if their productivity state falls outside of φ. An operational firm that is under attack, may also no longer find it worthwhile to stay in the market. Therefore, there is a set $\varphi^{'}$ of productivities where firms find it worthwhile to remain in the market, defined by $\varphi' = \{\varphi : V^a(Z, \varphi) > 0\} \subset \varphi$ whereas firms whose productivity $\varphi \in \varphi \setminus \varphi'$ will exit when under attack.

If the mass of entrants in a given period is M, the total number of entrants who become operational equals $M \sum_{\varphi \in \varphi} g(\varphi)$. Denote the total number of firms in the market by $L = \sum_{\varphi \in \varphi} L(\varphi)$, where $L(\varphi)$ is the sum of firms that are safe or under attack in market φ. Denote by $s(Z, \varphi)$ the share of safe firms in market φ. A share σ of firms in the market in any period will exit exogenously. No safe firm in the market will voluntarily choose to exit. All firms under attack whose productivity $\varphi \in \varphi \setminus \varphi'$ will voluntarily exit the market. Additionally, the shares $s(Z, \varphi) \sum_{\varphi' \notin \varphi} \Pi(\varphi'|\varphi)$ and $(1 - s(Z, \varphi)) \sum_{\varphi' \notin \varphi'} \Pi(\varphi'|\varphi)$ will also voluntarily exit, the first are safe firms and the second are firms under attack that transition into non-profitable productivities. In a steady state equilibrium, the mass

[20] While in Hopenhayn (1992) there exists a lower bound φ^* above which all productivities are profitable, this is not necessarily the case here because mid-productivity firms may have lower expected profits than low-productivity firms if they fall victim to cyber attacks more frequently.

of firms in each operational market $\varphi \in \varphi'$ evolves according to

$$\begin{aligned} L(\varphi) \;=\; & \underbrace{(1-\sigma)L_{-1}(\varphi)}_{\text{no exogenous separation}} + \underbrace{Mg(\varphi)}_{\text{new entrants}} \\ & \underbrace{-(1-\sigma)L_{-1}(\varphi)\left(\sum_{\varphi' \neq \varphi} \Pi(\varphi|\varphi')\right)}_{\text{exit to } \varphi'} + \underbrace{(1-\sigma)L_{-1}(\varphi')\left(\sum_{\varphi' \neq \varphi} \Pi(\varphi'|\varphi)\right)}_{\text{entry from } \varphi'} \end{aligned}$$

while in market $\varphi \in \varphi \setminus \varphi'$ the total mass of exiting firms (the RHS) must equal the total mass of entering operational firms (the LHS)

$$M \sum_{\varphi \in \varphi} g(\varphi) = \sigma L + (1-\sigma) \sum_{\varphi \in \varphi \setminus \varphi'} (1 - s(\theta))\, L(\varphi)$$

Breaking this down by productivity market, the total entrants and exiters in market φ are

$$entrants(Z,\varphi) = \begin{cases} Mg(\varphi) & \text{if } \varphi \in \varphi \\ 0 & \text{otherwise} \end{cases} \quad exiters(Z,\varphi) = \begin{cases} \sigma L(\varphi) & \text{if } \varphi \in \varphi' \\ \sigma L(\varphi) + (1 - s(Z,\varphi))\, L(\varphi) & \text{if } \varphi \in \varphi \setminus \varphi' \\ 0 & \text{otherwise} \end{cases}$$

Equality of entrants and exiters in a steady state implies the mass of operational firms in market φ is

$$L(\varphi) = \begin{cases} \frac{Mg(\varphi)}{\sigma} & \text{if } \varphi \in \varphi' \\ \frac{Mg(\varphi)}{\sigma + (1 - s(Z,\varphi))} & \text{if } \varphi \in \varphi \setminus \varphi' \\ 0 & \text{otherwise} \end{cases}$$

Additionally, the share of safe firms is governed by the following law of motion in steady state. If $\varphi \in \varphi \setminus \varphi'$ then $s(Z,\varphi) = 1$ since all firms under attack will voluntarily exit the market. If $\varphi \in \varphi'$ the share of safe firms evolves according to $(1 - s'(\varphi)) = (1 - \delta)(1-\sigma)(1 - s(Z,\varphi)) + s(Z,\varphi)(1-\sigma)q(\varphi)\Lambda(a^*(\varphi), x^*(\varphi))$ where s' is the share of safe firms in the next period. Because in steady state $s' = s$, we have

$$s(Z,\varphi)\,|_{\varphi \in \varphi'} = \frac{1 - (1-\delta)(1-\sigma)}{1 - (1-\delta)(1-\sigma) + (1-\sigma)q(\varphi)\,\Lambda(a^*(\varphi), x^*(\varphi))}$$

3.2 Attackers

There is an unbounded set of cyber attackers who start out as *vacant*, that is they are not engaged in an attack. Vacant attackers can meet firms, which will lead to either a successful strike or a failed strike.

Vacant attackers. Vacant attackers must pay a fixed cost κ to acquire the necessary computational resources to strike a firm. Additionally, attackers must choose which firms φ to strike. An attacker who chooses firm φ will meet a firm of productivity φ in the next period with probability $\lambda(Z, \varphi)$. The value of a vacant attacker targeting firms φ is

$$W^v(Z, \varphi) = -\kappa + \{\lambda(Z, \varphi) W^m(Z, \varphi, x^*(Z, \varphi)) + (1 - \lambda(Z, \varphi)) W^v(Z, \varphi)\} \tag{9}$$

where $W^m(Z, \varphi, x^*(Z, \varphi))$ is the value derived from meeting a firm whose optimal cyber security investment is $x^*(Z, \varphi)$.

Active attackers. When an attacker meets a firm it chooses the attack intensity a with which to strike the firm. The value of meeting a firm of productivity φ and with cyber security investment x is

$$W^m(Z, \varphi, x) = \max_{a \in [0, A]} -a + \Lambda(a, x) W^a(Z, \varphi) + [1 - \Lambda(Z, a, x)] W^v(Z, \varphi) \tag{10}$$

where a translates into attack intensity $\mathcal{A}(a)$, which we assume is strictly increasing and concave. Note that the attacker chooses the resources to devote to the attack *after* observing the cyber security level of the firm. The FOC that pins down the attack intensity a upon meeting a firm with cyber security x is

$$\frac{\partial \Lambda(a, x)}{\partial a} [W^a(Z, \varphi) - W^v(Z, \varphi)] = 1 + \mu_A - \mu_0 \tag{11}$$

where μ_0 and μ_A are the Lagrange multipliers for the constraints $a \geq 0$ and $a \leq A$. Note that only one of these constraints can bind at a time.

If the attack is successful, the attacker gains value

$$W^a(Z, \varphi) = \chi \cdot Z\varphi \cdot (n^a)^\alpha + \beta \{(1 - \sigma) [(1 - \delta) W^a(Z, \varphi) + \delta W^v(Z, \varphi)] + \sigma W^v(Z, \varphi)\} \tag{12}$$

where χ is the share of output the cyber attacker is able to retain.

Successful Attacks. When a firm with cyber security x meets an attacker with attack intensity a, they respectively draw cyber shocks ϵ_f, ϵ_a from a Gumbell distribution with scale parameter ϱ. An attack is *successful* if $\mathcal{A}(a) + \epsilon_a \geq C(x) + \epsilon_f$ for a function $\mathcal{A}(\cdot)$ which represents the realized attack given intensity a and so the probability of a successful attack is

$$\Lambda(a, x) \equiv \Pr\left[C(x) + \epsilon_f \leq \mathcal{A}(a) + \epsilon_a\right] = \frac{\exp\left\{\frac{\mathcal{A}(a) - C(x)}{\varrho}\right\}}{1 + \exp\left\{\frac{\mathcal{A}(a) - C(x)}{\varrho}\right\}} \tag{13}$$

Note that the probability of an attack succeeding depends only on the relative levels of attack intensity a and cyber hiring x. However, in equilibrium, both x and a are contingent on firm productivity φ, implying that the probability of a successful attack varies across firm productivity levels.

3.3 Cyber Externalities

We introduce aggregate productivity spillovers arising from cybersecurity risk through the term Z, which affects firm-level production. This term captures the idea that cyberattacks generate negative externalities that extend beyond directly targeted firms. Through interconnected supply chains, shared payment systems, common software dependencies, and other forms of digital linkages, even firms that are not attacked can suffer disruptions or productivity losses.[21].

Analogous to the positive knowledge spillovers in Romer (1986, 1990), where aggregate productivity rises with the economy-wide stock of knowledge, we model cyber risk as generating a negative externality: aggregate productivity declines as a larger share of firms are compromised by a cyber attack. Formally, we specify

$$Z = \left(\int_{\varphi} (1 - s(Z, \varphi)), d\varphi\right)^{\epsilon_{\text{att}}} \tag{14}$$

where $s(Z, \varphi)$ denotes the share of firms of type φ currently under attack. The higher the economy-wide prevalence of attacks, the greater the drag on overall productivity, even for firms that remain unbreached.

[21]It is commonplace in the macro literature to have shocks to individual firms propagate through networks consistent with the network amplification mechanisms emphasized by Acemoglu et al. (2012)

3.4 Household

There is a representative household that derives utility by consuming the homogeneous good, earns wage w by providing a unit of labor inelastically, and earns firm profits. Its welfare is defined by

$$\mathcal{W} = \log(C) \tag{15}$$

where the aggregate resource constraint is $C = w + T + \Pi - f_e M \int_\varphi g(\varphi)$. Consumption is financed via labor income net of transfers[22] T, and firm profits net of entry costs, where profits are defined as

$$\Pi = \int_\varphi \Big\{ s(Z,\varphi) \left[\pi(Z,\varphi;w) - wx(\varphi) \right] \tag{16}$$

$$+(1-s(Z,\varphi))(1-\ell)\pi(Z,\varphi;w) \Big\} D(\varphi) d\varphi \tag{17}$$

We assume cyber agents are foreigners and therefore exclude the real resources paid to acquire attack intensity and the amount earned by attackers from the domestic resource constraint[23].

3.5 Equilibrium

Matching Technology. Let $s(Z,\varphi)$ be the share of firms of productivity φ that are safe from an attack and let $v(\varphi)$ be the vacancy rate of cyber attackers in market φ, defined as the number of vacant attackers per firm. If the mass of firms operating in market φ is $L(\varphi)$, only $s(Z,\varphi)L(\varphi)$ safe firms and $v(\varphi)L(\varphi)$ cyber vacancies give rise to new cyber attacks. The number of cyber attacks is $m\left(s(Z,\varphi)L(\varphi), v(\varphi)L(\varphi)\right)$, which is increasing in both arguments, concave, and homogeneous of degree 1. Defining cyber tightness as $\theta = \frac{s}{v}$, the vacancy-filling and the attack rates are

$$q\left(Z,\theta(Z,\varphi)\right) = m\left(\frac{s(Z,\varphi)}{v(Z,\varphi)}, 1\right) \tag{18}$$

$$\lambda\left(Z,\theta(Z,\varphi)\right) = m\left(1, \frac{v(Z,\varphi)}{s(Z,\varphi)}\right) = \theta(Z,\varphi) q\left(Z,\theta(Z,\varphi)\right) \tag{19}$$

[22] In the baseline, transfers are zero. We introduce them here anticipating our discussion of policies a government could undertake to reduce cyber risk in section 6.

[23] The Council on Foreign Relations estimates that 77% of attacks on U.S. firms originated in five of its geopolitical rivals.

Steady State Equilibrium. The steady state equilibrium is a collection

$$Z, w, M, \left\{ x(\varphi), a(\varphi), n(\varphi), \theta(\varphi), s(Z,\varphi), \Lambda(\varphi), \theta(\varphi), \lambda(\varphi), q(\varphi) \right\}_{\varphi \in \varphi}$$

such that

1. Given w, firms choose $n(\varphi)$ to maximize their static profits and $x(\varphi)$ is chosen to solve their dynamic problem.
2. Taking the firm choices of security $x(\varphi)$ as given, attackers' choice of $a(\varphi)$ solve their dynamic problem.
3. Firms' free-entry condition holds:

$$\sum_{\varphi \in \varphi} V^e(\varphi)\, g(\varphi) - f_e = 0$$

4. The probability of a successful attack is consistent with choices made by the firms and attackers in each submarket,

$$\Lambda(\varphi) = \frac{\exp\left\{ \frac{\mathcal{A}(a(\varphi)) - C(x(\varphi))}{\varrho} \right\}}{1 + \exp\left\{ \frac{\mathcal{A}(a(\varphi)) - C(x(\varphi))}{\varrho} \right\}}$$

5. Free entry for attackers targeting each productivity φ holds:

$$W^v(\varphi) = 0$$

6. The labor market clears

$$1 = \sum_{\varphi \in \varphi'} n^a(\varphi)(1 - s(Z,\varphi))L(\varphi) + \sum_{\varphi \in \varphi} n(\varphi) s(Z,\varphi) L(\varphi)$$

   where the first summation is the labor demand coming from safe and attacked firms when endogenous exit does not occur and the second summation is the labor demand coming from safe firms only because attacked firms endogenously exit.
7. The externality in equation (14) holds.
8. Cyber tightness and the contact rates for firms and attackers are consistent with the matching function in the cyber market.

3.6 Discussion of Model Assumptions

One of this paper's main contributions is the development of a first macroeconomic, general equilibrium model to analyze cyber attacks and study policy related to attacks. In this subsection, we discuss some key modelling decisions and how they map to salient features that underpin the relationship between cyber attackers and the firms they strike.

Assumption #1: Stackelberg Structure. In our model, firms pre-commit to a level of defense, acting as Stackelberg leaders. Our modeling approach is standard in the literature on security games[24] and is closest in spirit to Ebel and Mitra (2024), and builds on the intuition that firm-level security investments are ongoing investments in generally maintaining a level of security which is broadly observable. For instance, firms may actively disclose investments in cyber security in annual reports to investors (Al Amosh and Khatib 2025). The key question in interpreting this assumption is whether the typical attacker can infer the cybersecurity level at a prospective target based on observables like firm performance and size with a high degree of precision; we argue that this is eminently plausible. Direct evidence for such a mechanism would require a systematic forensic analysis of cyberattackers and their information sets.

Assumption #2: Search, Matching and Randomness in Attack Success. We model two layers of randomness in the rate at which attackers and firms meet: first, there is uncertainty due to the search framework in which attacks on firms that are more likely to be targeted are less likely to meet the firm at all, and second, conditional on meeting, attacks may fail even if the realised intensity of an attack exceeds the security level of the firm. We now discuss both layers.

The first layer of uncertainty, driven by imperfect matching, reflects the fact that targeting the most attractive targets is subject to congestion. Intuitively, we wish to model the idea that attackers face congestion frictions in attacking the largest firms. This congestion arises because at any given point in time, attackers who successfully breach a firm crowd out other attackers in three possible ways. First, ransoming a firm's data may physically prevent access to other potential attackers by encrypting files and rendering systems or data inaccessible (Microsoft 2023; National Cyber Security Centre 2023). Second, data compromised by a first attacker lowers the marginal return to additional attackers because the target's systems are already disrupted or partially monetized (though repeat

[24]This assumption is also standard in the game theory literature on security investments more broadly (Korzhyk et al. 2011), which is based on the idea that attackers constitute rational agents able to observe ongoing security investments and activities. In the cybersecurity context, a recent piece extending the analysis of Stackelberg security games is Huang, Naghizadeh and Liu (2024).

attacks can still occur Federal Bureau of Investigation and CISA 2023). Third, attacked firms typically respond by isolating systems, reducing exposure, and patching vulnerabilities, making them less likely targets and hence less attractive targets for subsequent attackers (Cybersecurity and Infrastructure Security Agency 2025).

The second layer of uncertainty is that conditional on matching, attackers may fail to breach a firm's defenses stochastically. We do this because whether or not an attack succeeds depends both on the decisions of the parties at play – firms choose how much to invest on their defenses and attackers choose which firms and how intensely to attack them – but also from randomness such as individuals clicking malicious links (Verizon 2024; National Institute of Standards and Technology 2023). Further, we note that attacks on firm systems often fail to breach all the defenses they seek to overcome; rather, attackers are often caught and blocked rapidly, a factor that Ponemon Institute and IBM (2025) argues is responsible for the minor decline in losses from a typical attack.

Assumption #3: Spillovers. We model spillovers as productivity losses which are a concave function of the share of firms under attack: as the share under attack grows, the rate at which spillover-driven losses rise increases. Our motivation for this is an extensive literature documenting that cyberattacks generate losses via network effects through supply chains (Acemoglu, Malekian and Ozdaglar 2016) or through common choices for security vendors (Anand, Duley and Gai 2026), and that losses from densely connected networks are initially convex in the share of affected nodes. Extending our work to include explicit network-driven spillovers and subsequent losses is an important area for future research, and will require much more data to adequately discipline.

4 Calibration

We calibrate the model in two stages. First, we discipline it to a "no-cyber" steady state designed to approximate the U.S. economy around the year 2000, when firms were largely unaffected by cyber threats. Second, we re-calibrate the model to match key features of the U.S. economy in the 2020s, a period characterized by the pervasive presence of cyberattacks.

4.1 "No cyber" calibration

We choose the 2000 baseline to represent a period in which firms were essentially not subject to cyber attacks[25]. To do so, we set the share of output lost by firms and the corresponding gain accruing to attackers during a successful cyber incident to zero, ensuring that cyber activity is absent in equilibrium. With these restrictions, no firm has an incentive to incur the cost of cybersecurity investment, and no attacker has an incentive to pay the fixed cost of entry.

The core technology and discounting parameters are set externally. The degree of decreasing returns is $\alpha = 0.85$, in line with the estimate in Ottonello and Winberry (2020), and the discount factor is $\beta = 0.96$, targeting a 4 percent annual interest rate. Total labor supply is normalized to one. The remaining parameters are estimated via simulated method of moments to match three salient features of the U.S. firm distribution in 2000: (i) the distribution of entrants, (ii) the overall distribution of firm sizes, and (iii) the average exit probability of firms. This yields estimates for the shape of the Pareto productivity distribution at entry, ϖ, the exogenous exit probability σ, the fixed costs of entry and operation (f_e and f), and the parameters governing the log-productivity process, $(\mu_z, \rho_z, \sigma_z)$. Table 4 reports the resulting parameter values.

The implied firm and entrant size distributions are shown in panels A and B of Figure 6. The model closely matches the empirical distributions and generates an average exit probability of 6.0%, compared to the observed 6.2% from Lee and Mukoyama (2015).

4.2 Cyber calibration

Holding fixed the parameters estimated in the "no cyber" economy, we calibrate the additional parameters that govern cyber risk. The escape probability, δ, is set to match the half-life of the decline in firm sales following a cyber attack. We normalize the EV scale parameter to $\varrho = 1$, the maximum attack capacity to $A = 1$, and the cost-scaling parameter of the attack function to $\kappa_A = 1$.

The remaining parameters are estimated via simulated method of moments to match four sets of moments of the cyber economy in the 2020s: (i) the average cybersecurity employment share by firm size in 2024, (ii) total cyber losses equivalent to 1.16 percent

[25] In 2000, digital penetration in the economy was still in its nascent stages, with internet access being highly limited. Only 6% of the world population was online in 2000, compared to 68% by 2025. As of March 2001, only 6.3% of US firms were procuring inputs online (OECD 2001), and in 1999, sales over the internet amounted to 0.6% of total sales (while Canada was at 0.2%) (StatCan 2000). Accordingly, cyberattacks in this period were largely conducted for notoriety, and limited digitalisation also limited losses from such attacks(ICAEW Insights 2023).

of GDP, (iii) the share of firms experiencing a cyber incident (0.67 percent), and (iv) an amplification factor linking direct to total economic losses from cyber shocks equal to 4.

The empirical targets underlying these moments are drawn from a combination of micro and macro sources. The first moment, the average cybersecurity employment share by firm size in 2024, has been discussed in detail earlier in the paper. The second moment, total cyber losses equal to 1.16 percent of GDP, is based on macroeconomic estimates of cyber losses reported in the International Monetary Fund's Global Financial Stability Report (International Monetary Fund 2023). The third moment, the share of firms experiencing a cyber incident of 0.67 percent, is computed using incident-level loss data from Coveware as reported via Bloomberg, combined with the firm size distribution from the U.S. Business Dynamics Statistics to map observed incidents into an economy-wide firm-level attack probability. Finally, the amplification factor mapping direct to total losses is taken from Crosignani, Macchiavelli and Silva (2023), who document that propagation through supply chains raises total economic losses to roughly four times the initial direct loss.

The estimated parameters are shown in table 4. The loss and gain shares, $\ell = \chi = 0.275$, are assumed to be equal so that the resource constraint is satisfied without any further adjustments.[26] The matching function is constant returns to scale, $m(\theta) = \psi\theta^{\eta}$, with $\psi = 0.01$ and $\eta = 0.35$. The attack function is $K(a) = \kappa_A \frac{a^{1-\gamma_A}}{1-\gamma_A}$ with $\gamma_A = 0.2$, and the cybersecurity function is $C(x) = \kappa_C \frac{x^{1-\gamma_C}}{1-\gamma_C}$ with $\kappa_C = 0.25$ and $\gamma_C = 0.825$. The attacker vacancy posting cost is $\kappa_v = 0.0002$, and the externality parameter is 1.0001.

The calibrated model delivers a loss-to-output ratio of 0.92% (against the empirical 1.16%), a share of firms under attack of 0.62% (vs. 0.67% in the data), and a ratio of indirect to direct losses of 3.94 (vs. 4.0 taken from Crosignani, Macchiavelli and Silva 2023). Panel C of figure 6 compares the model-predicted cybersecurity employment shares by firm size with data from Revelio Labs (via LinkedIn). The model replicates the increasing investment share in firm size, with large firms allocating 0.44% of labor to cybersecurity compared to 0.46% in the data.

The aggregate consequences of introducing cyber risk into the economy are shown in Table 5. Through the propagation externality, cyber attacks reduce aggregate total factor productivity by 0.62%. Lower productivity, together with the risk of being attacked and losing revenue, reduces the value of entry for firms, leading to a 3.61% decline in the mass of entrants M. As a result of these forces, the introduction of cyber risk lowers aggregate output by 1.76% and consumption by 1.66%.

These results resonate with the mechanism discussed in Baksy, Caratelli and Olson

[26] This assumption is conservative: evidence suggests that firm losses can exceed attacker gains (Anderson et al. 2013).

	Parameter (quarterly frequency)	Value
β	Discount factor	0.96
α	Returns to scale	0.85
ϖ	Pareto distribution scale	0.7
σ	Exit probability (exogenous)	0.00
f	Per-period fixed cost of production	0.183
f_e	Fixed cost of firm entry	3.83
ρ_z	Persistence in productivity	0.925
σ_z	Stdv. of productivity innovations	0.5
ℓ	Firm loss share	0.275
χ	Attacker gain share	0.275
δ	Escape probability	1
η	Elasticity of matching function	0.35
ψ	Scaling matching function	0.01
$K(a)$	Cost of attack function	$\frac{a^{1-0.2}}{1-0.2}$
$C(x)$	Cost of security function	$0.25 \cdot \frac{x^{1-0.825}}{1-0.825}$
κ_v	Elasticity of matching function	0.0002
ϵ_{att}	Romer externality	1.0001

Table 4: Model parameters. The first set of parameters is set by the authors to match an annual interest rate of 4% and to match the decreasing returns in Ottonello and Winberry (2020). The second set of parameters is estimated via SMM to match key moments from the 2000 ("no cyber") U.S. economy. The third set of parameters is estimated via SMM to match cyber-related moments from the 2020 ("cyber") U.S. economy.

	Productivity (Z)	Entry Mass (M)	Output (y)	Consumption (C)
2000	1.0000	0.00370	0.4862	0.4593
2020	0.9938	0.00357	0.4776	0.4517
Percent change (%)	-0.62	-3.61	-1.76	-1.66

Table 5: Aggregate Outcomes in 2000 and 2020

(2025). Small firms face few attacks because they offer little reward to attackers. Large firms are attractive targets but offset this risk through extensive cybersecurity defense. In contrast, mid-sized firms occupy a vulnerable middle ground: valuable enough to attract attackers but too resource-constrained to sustain comprehensive defenses.

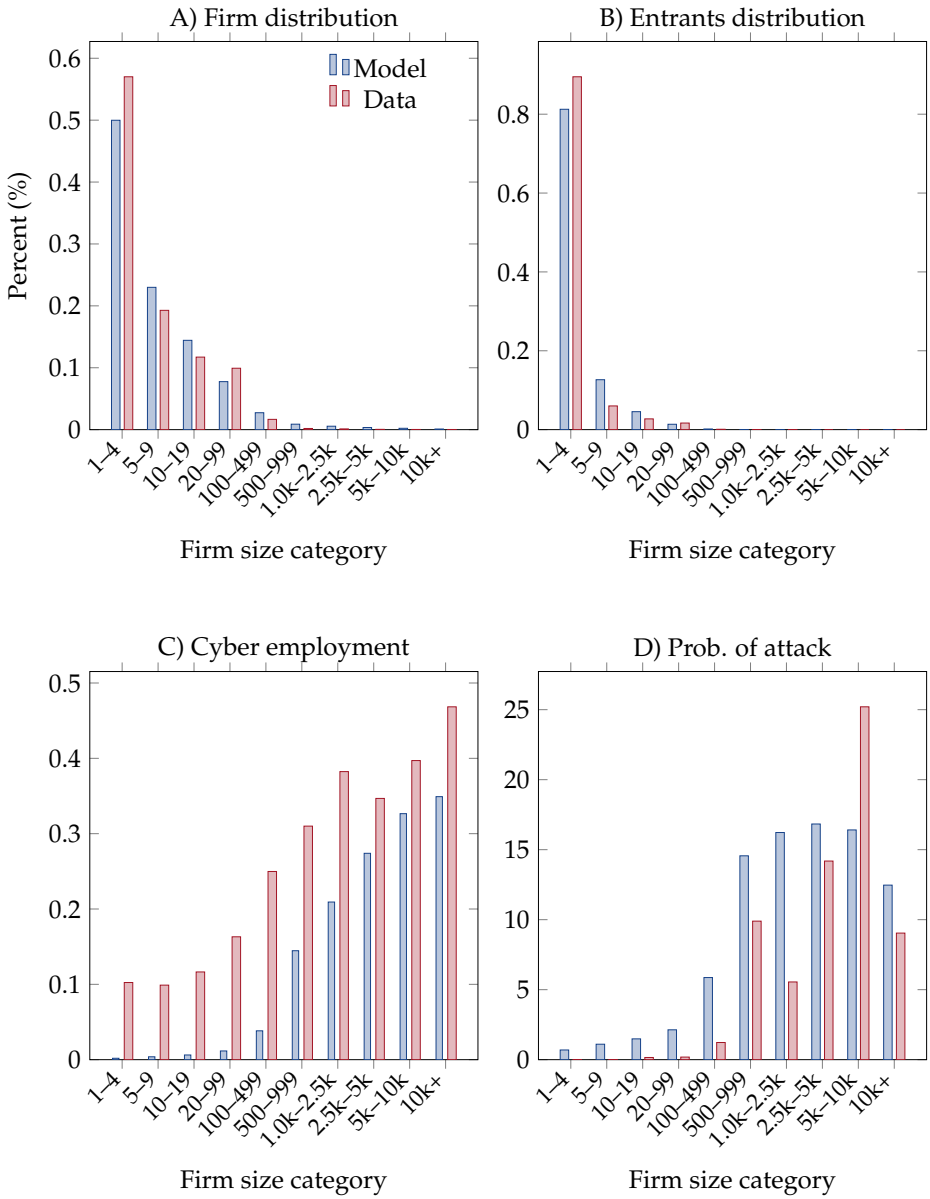


Figure 6: Panel A shows the firm distribution by firm size in the data (red) and the model (blue). Panel B shows the distribution of entering firms by firm size in the data (red) and the model (blue). Empirical distribution reflects the year 2000. Model represent the calibrated model without cyber-attacks. Panel C shows the share of employment devoted to cyber by firm size. Panel D shows the probability of attack by firm size. Source: Bloomberg via Coveware, Compustat, and authors' analysis.

4.3 Validation

Panel D of figure 6 shows that the model also qualitatively captures the varying annual probability of attack of firms of different sizes – an untargeted moment. The probability of a firm being attacked peaks for mid-sized firms both in the model and in the available data from Coveware. These two last pieces of evidence establish that larger firms exhibit stronger cyber preparedness than smaller firms and that medium-sized are those most susceptible to attack.

To understand the source of the inverse-U pattern in cyber risk we return to the theoretical model and show that mild conditions lead to this result. Holding fixed aggregate productivity Z, the wage w, and the firm-side contact probability q, we isolate the partial-equilibrium mechanism operating through firms' security choices and attackers' intensity choices. The object of interest is therefore the contact-conditional probability that a matched attack succeeds:

$$\Lambda(\varphi) \equiv \Lambda(a^*(\varphi), x^*(\varphi)) = \frac{\exp\left\{\frac{\mathcal{A}(a^*(\varphi)) - C(x^*(\varphi))}{\varrho}\right\}}{1 + \exp\left\{\frac{\mathcal{A}(a^*(\varphi)) - C(x^*(\varphi))}{\varrho}\right\}}.$$

Because labor demand and output are strictly increasing in productivity, the comparative statics in φ map directly into firm size. Since q is held fixed in this exercise, the same shape also applies to the overall probability of a successful attack up to the constant factor q.

Lemma 1 (Monotone cybersecurity investment)**.** In partial equilibrium, $x^*(\varphi)$ is strictly increasing in φ.

Proof. See appendix A. □

This monotonicity result is the first step in the argument. Higher-productivity firms are larger and have more to lose from disruption, so they defend more aggressively ex-ante.

Proposition 1 (Hump-shaped cyber incidence)**.** Suppose that $\mathcal{A}'(a) > 0$, $C'(x) > 0$, and that the attacker's second-order condition holds whenever the attack-intensity choice is interior. Suppose further that the following two conditions hold.

First, in the lower tail of the productivity distribution, cybersecurity investment is only weakly responsive to productivity. That is, there exist $\underline{\varphi}$ and $\bar{m}_x \geq 0$ such that

$$0 \leq x^{*\prime}(\varphi) \leq \bar{m}_x \qquad \text{for all } \varphi \leq \underline{\varphi},$$

where $\bar{m}_x$ is sufficiently small that

$$C'(x^*(\varphi))\bar{m}_x < \mathcal{A}'(a^*(\varphi))a^{*\prime}(\varphi) \qquad \text{for all } \varphi \leq \underline{\varphi}.$$

Second, in the upper tail of the productivity distribution, the attacker's capacity constraint binds, that is there exists $\bar{\varphi}$ such that

$$a^*(\varphi) = A \qquad \text{for all } \varphi \geq \bar{\varphi}.$$

Then the probability of a successful attack is hump-shaped in firm productivity: it is increasing for sufficiently small φ and decreasing for sufficiently large φ.

Proof. See appendix A. □

The first condition of the proposition formalizes the idea that, among small firms, cybersecurity investment does not yet scale strongly with firm size. This assumption is motivated by the empirical size gradient in cybersecurity employment documented in Figure 4: cyber employment shares are low and relatively flat among the smallest size bins, while the steep increase in cybersecurity intensity occurs higher in the firm-size distribution. Because firm employment is increasing in productivity in the model, the observed size-gradient in cybersecurity employment provides direct evidence on the shape of $x^*(\varphi)$ across the productivity distribution. In the lower tail, moving to a larger firm primarily raises the attacker's prize, while the firm's additional defensive investment remains limited. In the upper tail, by contrast, large firms invest heavily in cybersecurity and attackers eventually reach their capacity constraint. The combination of these two forces generates the inverse-U shape.

The economic interpretation is straightforward. Small firms are not sufficiently valuable to justify intense attacks. Large firms are valuable, but they also invest heavily in cybersecurity, and eventually force attackers against their capacity limit. Intermediate firms lie between these two margins: they are valuable enough to attract attackers, but not yet protected enough to neutralize them. That is why successful-attack risk is highest in the middle of the firm distribution.

5 Counterfactual Exercises

Cyber risk has grown over the past two decades and is likely to intensify as firms accumulate sensitive data, become more digitally dependent, and as technologies such as artificial intelligence enhance the effectiveness of cyber attackers. To quantify how these

structural shifts may affect equilibrium outcomes, we study two counterfactual scenarios. The first increases attackers' capacity, A, while the second raises the probability that firms are successfully targeted. Both exercises capture plausible forms of technological progress that favor cyber agents. Together, they show that evaluating cyber risk requires accounting for endogenous firm responses and general equilibrium feedbacks.

For each counterfactual, we compare the steady state of the economy to the benchmark equilibrium and express the resulting change in aggregate output as a percent deviation from the benchmark. We decompose the change in aggregate output into three components. Let Y denote equilibrium aggregate output. Direct losses are measured as

$$L^{\text{direct}} \equiv \ell Z \sum_{\varphi \in \Phi'} \varphi \left(n^a(\varphi)\right)^{\alpha} L(\varphi)$$

which captures the loss in output from successful cyber attacks. Indirect losses are defined as

$$L^{\text{indirect}} \equiv W \sum_{\varphi \in \Phi} x(\varphi) L(\varphi) + \frac{1-Z}{Z} Y,$$

and capture the added costs due to cyber security investment and the lost output due to the cyber externality. The residual is given by

$$L^{\text{residual}} \equiv Y - L^{\text{direct}} - L^{\text{indirect}}.$$

The residual component reflects all endogenous general equilibrium responses, including adjustments in firm entry and in attack incidence.

Panels (A) and (C) report the output decomposition for changes in attack capacity and matching efficiency, respectively, while Panels (B) and (D) display the associated endogenous equilibrium responses underlying the residual component.

Higher attack capacity A. Panels (A) and (B) report the effects of increasing attacker capacity from $A = 1$ to $A = 5$. Aggregate output declines monotonically with attack capacity, falling by about 2.5 percent relative to the benchmark economy at $A = 5$. Panel (A) shows that only a limited share of this decline is attributed to cyber losses. Direct losses from successful attacks account for roughly 0.4 percentage points (about 16 percent of the total decline), while indirect losses associated with the propagation externality contribute an additional 0.07 percentage points. The remaining share of the output contraction—over four-fifths—is driven by endogenous equilibrium responses.

Panel (B) illustrates the mechanisms behind this residual component. As attack capacity increases, the mass of active firms declines sharply, reflecting weaker entry incentives

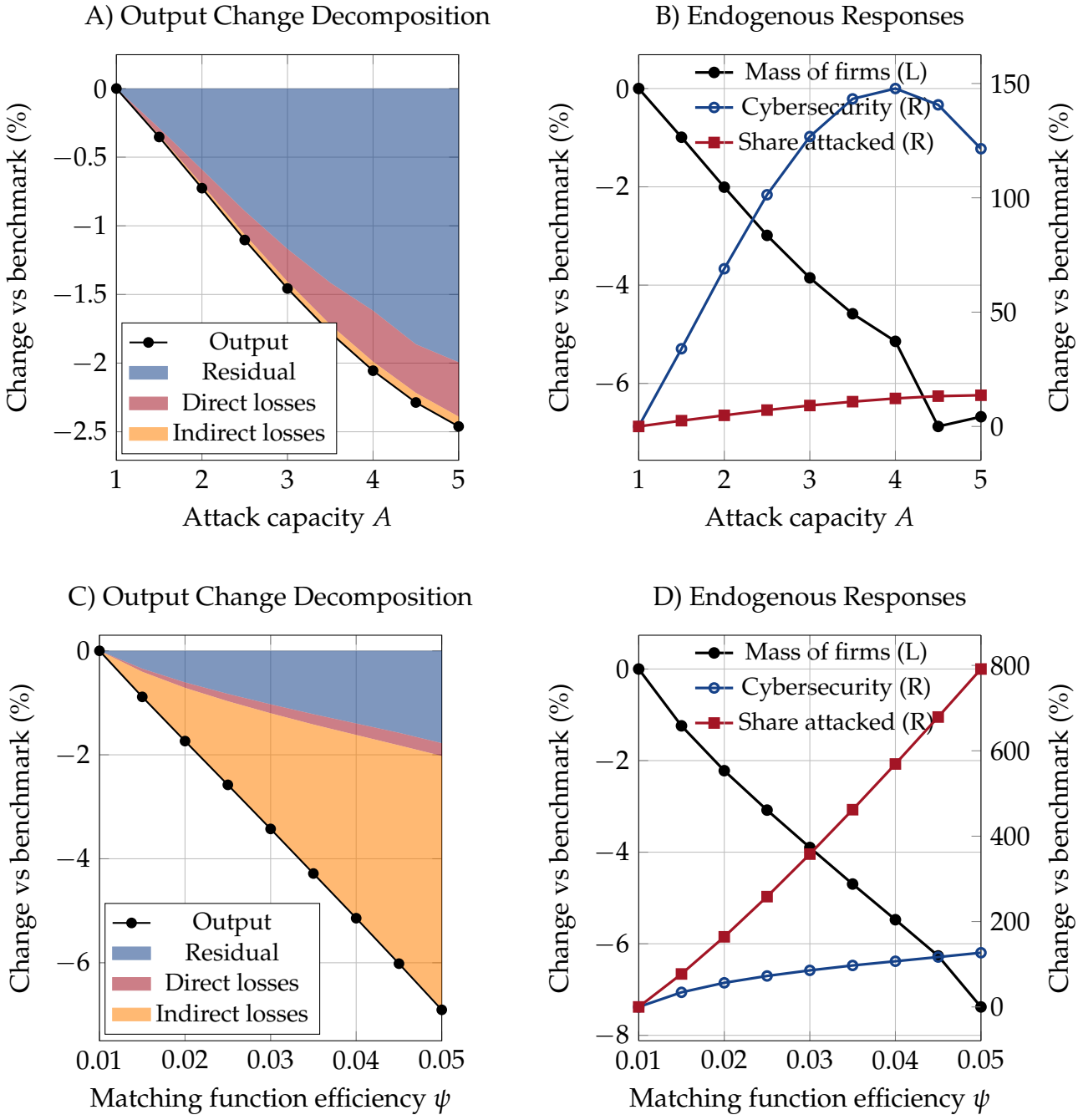


Figure 7: Output decomposition and endogenous responses to cyber risk and labor market efficiency. Panels (A) and (B) report the effects of higher attack capacity A. Panel (A) decomposes the percent change in aggregate output relative to the benchmark economy into direct losses from cyber attacks, indirect losses associated with the propagation externality, and a residual. Panel (B) shows the corresponding endogenous equilibrium responses: firm entry, cybersecurity investment, and the share of firms under attack. Panels (C) and (D) present the same objects for changes in matching function efficiency ψ.

in a riskier cyber environment. At the same time, surviving firms raise cybersecurity investment in response to heightened risk. Despite these defensive efforts, the share of firms that are successfully attacked rises substantially, highlighting the limits of individual firm responses when attack capacity increases. Together, reduced firm entry, higher defensive costs, and increased attack incidence generate large general equilibrium effects that dominate the direct and indirect loss channels.

Higher matching efficiency ψ. Panels (C) and (D) present the analogous decomposition for increases in cyber matching function efficiency ψ. As ψ rises, aggregate output declines sharply and is almost 7 percent lower when $\psi = 0.05$ compared to the benchmark of $\psi = 0.01$. Panel (C) shows that this decline is driven primarily by indirect losses associated with the propagation externality, which account for about 4.87 percentage points of the output reduction ($\sim$ 70% of the total). Direct losses from cyber attacks contribute an additional 0.245 percentage points ($\sim$ 3.6% of the total), while the remaining 1.78 percentage points ($\sim$ 26% of the total) are attributable to endogenous general equilibrium responses captured by the residual component.

Panel (D) highlights the underlying equilibrium adjustments. An increase in matching efficiency ψ has qualitatively similar effects to an increase in attack capacity in the previous counterfactual, but the quantitative implications are markedly different. In particular, the response of the share of firms under attack is substantially stronger when ψ rises. As Panel (D) shows, the fraction of firms experiencing a successful attack increases by almost a factor of eight as ψ moves from 0.01 to 0.05. By contrast, the same object rises by only about 13 percent when attack capacity increases from $A = 1$ to $A = 5$.

This stark difference explains why indirect losses play a much larger role in the ψ counterfactual. The sharp increase in attack incidence strongly depresses aggregate productivity through the cyber propagation externality, generating a large negative contribution of indirect losses to output. By comparison, changes in cybersecurity investment and in the mass of active firms are of similar magnitude across the two counterfactuals. Next, we look into why this channel is so much stronger when ψ rises than when A rises.

A versus ψ. The stronger role of indirect losses under increases in matching efficiency ψ reflects how attackers reallocate their effort across firm sizes. When attack capacity A rises, cyber attackers optimally concentrate attacks on large, high-output firms, which are more attractive targets. These large firms have a harder time keeping up their cyber investments when cyber attackers' A increases. As the panels in figure 8 show, attack success probabilities increase primarily at the top of the firm size distribution. Because

large firms represent only a small mass of firms, this shift in target by attackers limits the aggregate propagation externality, even though these firms are economically important.

By contrast, higher matching efficiency ψ raises the effectiveness of attacks across a much broader range of firm sizes. Attack success probabilities increase substantially not only for large firms but also for small and medium-sized firms, which are far more numerous. This expansion sharply amplifies the propagation externality and explains why indirect losses dominate the output response in the ψ counterfactual, despite similar aggregate changes in cybersecurity investment and firm entry.

Why do small firms not respond to the rise in attack risk with higher cybersecurity investments? The large increase in the share of attacked firms depresses TFP, which in our model depresses the gap $V^s(Z,\varphi) - V^a(Z,\varphi)$. When this gap between the values of safe and attacked firms is smaller, firms endogenously decide to invest less in cybersecurity because the marginal gain from being a safe firm is lower. However, in equilibrium, this further raises the share of firms under attack, feeding back and leading to a much lower responsiveness of cyber investments to cyber risk.

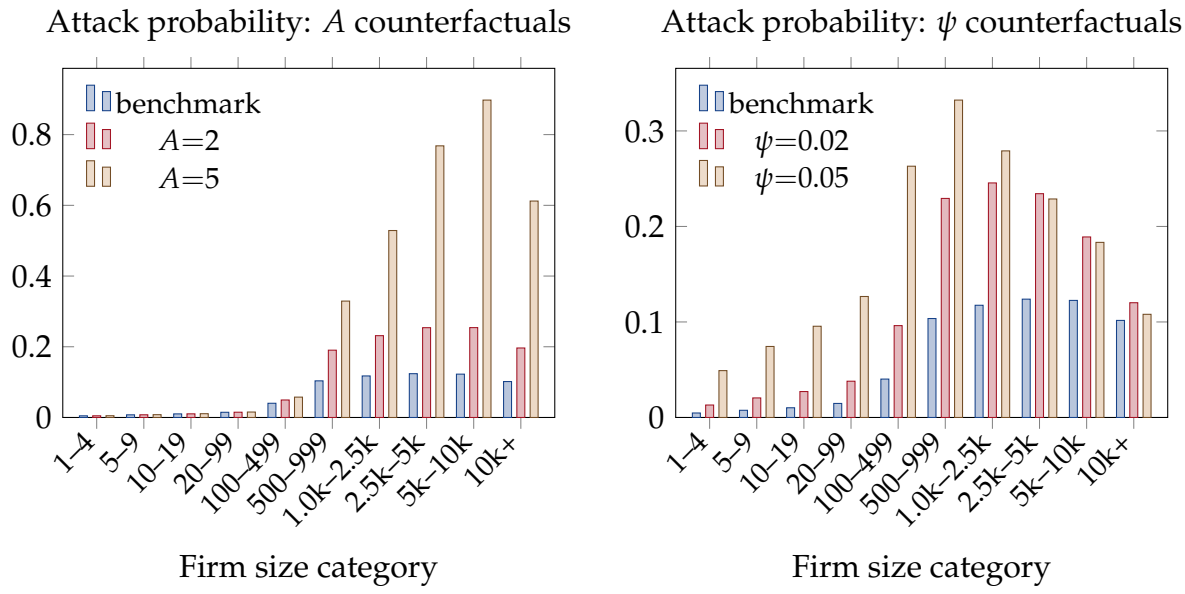


Figure 8: Attack success probability by firm size under changes in attacker capacity A (left) and matching efficiency ψ (right). Bars show the benchmark and two counterfactual values for each experiment.

These shifts across the firm size distribution translate into changes in market structure. Figure 9 reports the share of aggregate output produced by each firm size bin. Panel A varies attacker capacity A, while panel B varies matching efficiency ψ. The two adverse cyber scenarios have opposite implications for output concentration. As A rises,

large firms, being more exposed to successful attacks, lose output share, leading to a decline in concentration at the top of the size distribution. In contrast, higher ψ benefits, in relative terms, larger firms, which are better able to defend against attacks. As a result, output shifts away from small and mid-sized firms toward the largest firms. These patterns highlight how different dimensions of cyber risk can push market structure in opposite directions, an important consideration in economies where production is increasingly concentrated and potentially more vulnerable to systemic cyber events.

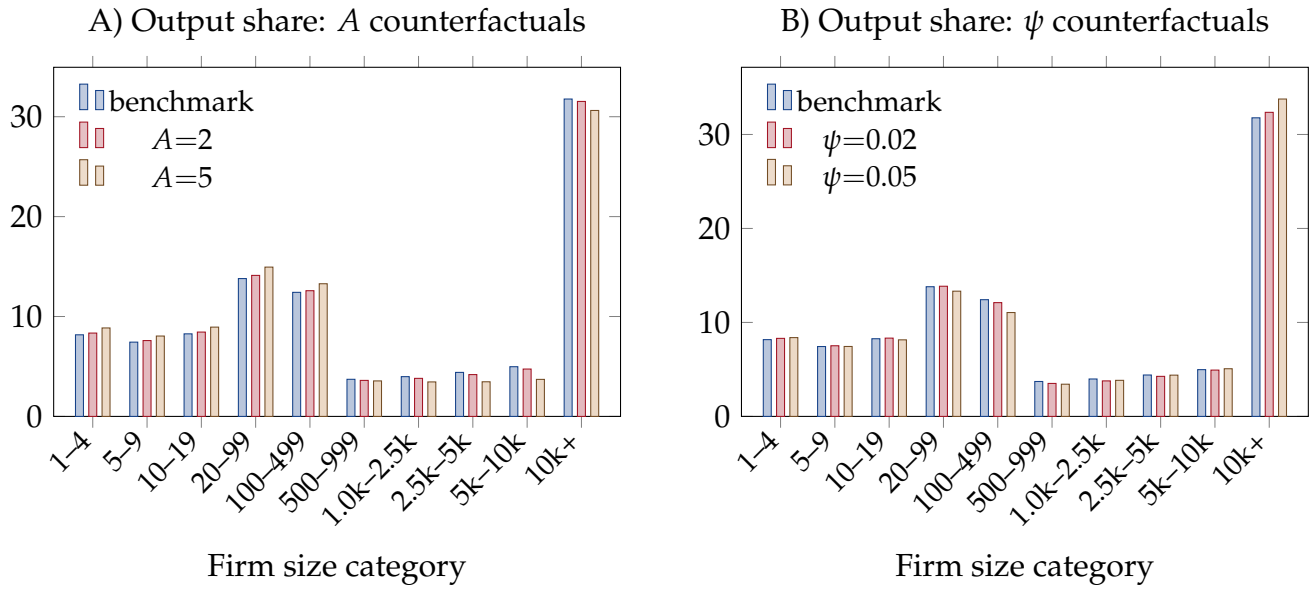


Figure 9: Share of output produced across the firm size distribution under changes in attacker capacity A (panel A) and matching efficiency ψ (panel B). Bars show the benchmark and two counterfactual values for each experiment.

6 Cyber Policy: Bailouts, Subsidies, and Regulation

In this section we examine the role that government policy can play in countering the rising tide of cyberattacks. We focus on two types of interventions: reactive policies, such as bailouts, that compensate firms after an attack occurs, and preventative policies, such as subsidies and regulatory requirements on cyber investment, that strengthen firms' defenses against attacks in the first place. While such measures may currently appear remote, large-scale and systemic cyber incidents may increasingly force governments to intervene. The UK government has already bailed out Jaguar following a major cyber attack through a £1.5 billion loan guarantee (UK Government, Department for Business

and Trade and UK Export Finance and HM Treasury 2025), and regulatory requirements, while vague and not particularly punishing, are already in place for some firms.[27]

The three instruments operate through different incentive channels. Subsidies and regulation act on firms' ex-ante cyber security choices, bailouts provide ex-post insurance against realized losses. We show that these policies differ sharply in how they affect firm entry, cybersecurity investment incentives, and the strength of the aggregate cyber externality. The remainder of the section makes these differences explicit both analytically and quantitatively.

6.1 Adding Policy to the Model.

Bailouts. We first analyze a bailout policy in which the government reimburses a share ι of losses suffered by attacked firms. In contrast to the subsidy scenario, bailouts leave the cost of cyber security unchanged but partially insure attacked firms by reducing the effective revenue loss from ℓ to $\ell(1-\iota)$. The Bellman equations become

$$\begin{aligned} V^s(\varphi) &= \max_{n,x\geq 0} \ \varphi n^\alpha - wn - wx - f \\ &\quad +\beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[q(\varphi')V^m(\varphi',x) + (1-q(\varphi'))V^s(\varphi')\right], \end{aligned} \tag{20}$$

$$\begin{aligned} V^a(\varphi) = \max\Big\{0, \quad & \max_n \left(1-\ell\cdot(1-\iota)\right)\varphi n^\alpha - wn - f \\ & +\beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}[(1-\delta)V^a(\varphi') + \delta V^s(\varphi')]\Big\}. \end{aligned} \tag{21}$$

The government finances these transfers with a lump sum tax T on the representative household.

$$T = \underbrace{\ell\cdot\iota\int_j \varphi n^\alpha(j)D^a(j)\,dj}_{\text{bailout payments}}. \tag{22}$$

Subsidies. Second, we consider a government that subsidizes firms' cyber security expenditures. The subsidy affects firm behavior through the marginal cost of cyber security investment x and via the distortions that the taxes required to fund the subsidy create.

Equations (2) and (5) become

$$V^s(\varphi) = \max_{n,x\geq 0} \ \varphi n^\alpha - wn - wx(1-\tau) - f$$

[27] See for example the FTC's Safeguards Rule.

$$+\beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[q(\varphi')V^m(\varphi',x)+(1-q(\varphi'))V^s(\varphi')\right], \tag{23}$$

$$V^a(\varphi)=\max\Big\{0,\quad \max_n(1-\ell)\varphi n^\alpha-wn-f \\ +\beta(1-\sigma)\mathbb{E}_{\varphi'|\varphi}\left[(1-\delta)V^a(\varphi')+\delta V^s(\varphi')\right]\Big\}. \tag{24}$$

The government taxes the representative household with a lump sum tax T and subsidizes a fraction τ of each firm's cyber security wage bill wx. Its budget constraint equates total tax receipts and subsidy expenditures so that the tax rate T adjusts to cover the government's cost of providing the subsidy.

$$T \;=\; \underbrace{\tau w\int_j x(j)D^s(j)\,dj}_{\text{subsidy outlays}}. \tag{25}$$

Regulation. We consider a third policy tool to combat cyber risk: minimum cybersecurity requirements. The government requires each firm with productivity φ to devote at least a fraction $\bar{r}(\varphi)$ of its total labor bill to cybersecurity. The requirement is progressive in firm productivity and is defined in odds form as

$$\rho(\varphi)=\frac{\bar{r}_{\text{reg}}}{1-\bar{r}_{\text{reg}}}\left(\frac{\varphi}{\varphi_{\text{reg}}}\right)^{\eta_{\text{reg}}},\qquad \bar{r}(\varphi)=\frac{\rho(\varphi)}{1+\rho(\varphi)}, \tag{26}$$

where $\bar{r}_{\text{ref}}\in[0,1)$ is the requirement for a firm with reference productivity $\varphi_{\text{reg}}>0$, and $\eta_{\text{req}}\geq 0$ governs progressivity. When $\eta_{\text{reg}}=0$, the requirement is uniform across firms: $\bar{r}(\varphi)=\bar{r}_{\text{ref}}$. When $\eta_{\text{reg}}>0$, more productive firms face a higher required cybersecurity share. Because cybersecurity and non-cyber labor are paid the same wage, the requirement is

$$\frac{x}{n+x}\geq\bar{r}(\varphi), \tag{27}$$

or, equivalently,

$$x\geq\rho(\varphi)n. \tag{28}$$

We assume that the regulation is mandatory and perfectly enforced, implying that firms cannot operate while violating this constraint. Accordingly, equation (2) becomes

$$V^s(Z,\varphi) \;=\; \max_{\substack{n,x\geq 0\\ x/(n+x)\geq\bar{r}(\varphi)}}\{Z\varphi n^\alpha-wn-wx-f \\ +\beta(1-\sigma)E_{\varphi'|\varphi}\left[q(Z',\varphi')V^m(Z',\varphi',x)+(1-q(Z',\varphi'))V^s(Z',\varphi')\right]\Big\} \tag{29}$$

Let (n^0, x^0) denote the firm's unconstrained choices. Some firms exhibit *voluntary compliance*: their unconstrained policies already satisfy

$$\frac{x^0}{n^0 + x^0} \geq \bar{r}(\varphi),$$

so the regulatory constraint does not bind. Other firms exhibit *regulatory compliance*: their unconstrained policies violate the requirement, and they therefore reoptimize employment and cybersecurity investment subject to the constraint. For these firms, the requirement binds:

$$\frac{x}{n + x} = \bar{r}(\varphi), \qquad \text{or equivalently} \qquad x = \rho(\varphi)n.$$

Finally, firms for which operating under the requirement is not profitable exit or do not enter.

6.2 Policy Impact

We now quantify how the three policies – bailouts, subsidies, and regulation – affect the economy, in general equilibrium. Partial equilibrium logic suggests that bailouts, a reactive policy, and subsidies and regulations, preventative policies, generate opposite investment incentives: bailouts reduce incentives to invest in cybersecurity, while subsidies and regulation increase them. While this intuition is broadly correct, it is only partly reflected once general equilibrium adjustments are taken into account.

We highlight three intertwined general-equilibrium channels: firm entry, the strength of the cyber propagation externality, and the distortionary effects of the policy (whether that is the fiscal burden as is the case of bailouts and subsidies or the (mis)allocation of labor). Figure 10 maps these channels by showing the effect that different subsidy and bailout rates have on aggregate economic indicators. The first column shows how various indicators evolve when varying the generosity of bailouts (red) and subsidies (blue), reflected in the fiscal burden to GDP of the policy on the x-axis. The second column shows the equivalent varying the minimum regulatory requirement, $\bar{r}_{\text{req}}$ on the x-axis and the progressivity of the requirement (from homogeneous $\eta_{\text{reg}} = 0$ to most progressive $\eta_{\text{reg}} = 1$). In each of these, the 0 rate case corresponds to the benchmark economy.

The policies have different consequences. The subsidy directly reduces the marginal cost of cyber security, inducing entering firms to invest a larger share of labor into cyber security. Higher investment improves resilience to cyberattacks, reducing the share of firms under attack and weakening the cyber externality, thereby raising economy-wide productivity. For moderate subsidy rates, the reduction in attack risk and the associated

weakening of the cyber externality dominate the distortionary effects of taxation, encouraging entry and increasing the mass of firms in the economy. However, as the subsidy becomes too generous ($\tau > 0.7$), the tax burden becomes increasingly onerous and more and more labor is directed towards security rather than production. While aggregate productivity continues to rise, firm entry flattens and later declines, ultimately reducing aggregate output. The model finds that, with respect to the current cyber environment, a generous subsidy covering 75% of cybersecurity investments is output and welfare-maximizing, increasing consumption by 0.13%.

Regulation, shown in the second column, affects the economy in a qualitatively similar way to subsidies. Regulation is also a preventative policy, forcing firms to reinforce their cyber defenses, leading to higher cyber investments, fewer attacked firms, and higher aggregate productivity. However, the effects of regulation vary widely depending on the progressivity of the requirements. For low minimum cyber requirements, progressivity does not affect the equilibrium very much. In fact, the output-maximizing regulatory policy is $\eta_{\text{reg}} = 0$ and $\bar{r}_{\text{req}} = 0.2\%$, which leads to an increase in output of 0.14% relative to the benchmark. However, as the minimum requirement increases, raising progressivity forces larger firms to devote more and more labor to their cyber defenses. But large firms are already well protected, so the impact on the share of firms attacked and on aggregate productivity is minimal. Therefore, higher progressivity leads to labor misallocation, as large firms must direct labor towards cyber security rather than production, without significant improvements in productivity. In turn, this lowers the value of entry for firms and therefore lowers the mass of firms in the economy. The misallocation of labor, coupled with the lower entry, leads to declining output for high values of progressivity, as the cyber requirement increases.

The effects of bailouts operate through a different channel. Because this is a reactive policy, even at relatively low bailout rates ι, bailouts weaken incentives to invest in cybersecurity through moral hazard. Lower investment raises attack incidence and strengthens the propagation externality leading to lower productivity. In addition, the increased tax burden discourages firm entry and as a result, aggregate output falls monotonically as the bailout rate increases.

Taken together, these results highlight that policies targeting ex-ante prevention and ex-post reaction differ sharply, especially once general equilibrium forces are taken into account.

State-dependent policies. The policy evaluation above suggests that subsidies and regulation can be effective at combating cyber attacks and increasing output. However, as

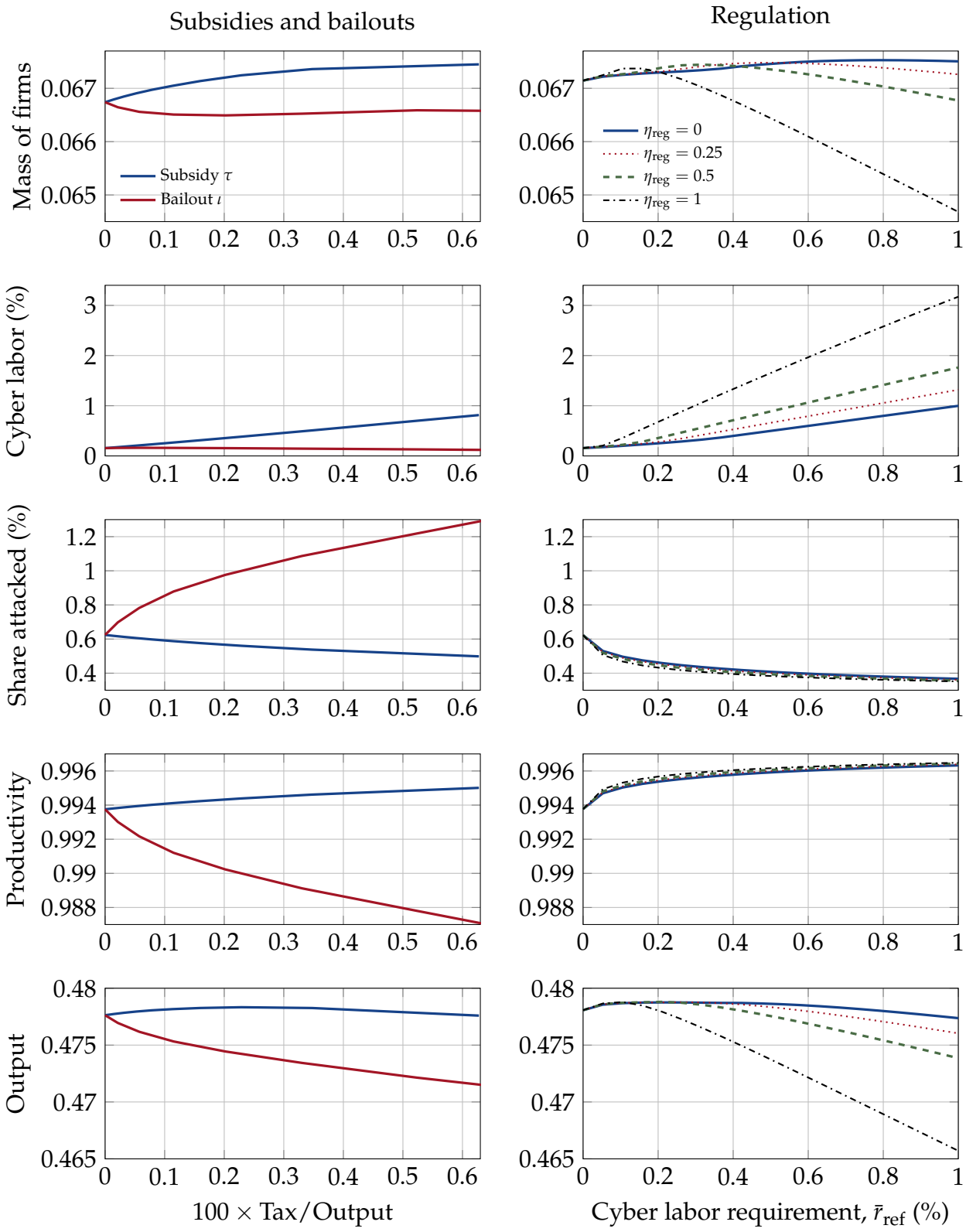


Figure 10: Effects of cybersecurity subsidies, bailouts, and minimum regulatory requirements on aggregate outcomes. The left column plots subsidies and bailouts against the associated tax-to-output ratio; the right column plots regulation against the minimum cyber labor requirement, for different degrees of progressivity.

cyber risk grows, the policy prescription may change. In figure 11 we consider the adverse cyber scenarios from the previous section. Panel A shows how output varies as the subsidy becomes more generous for the benchmark calibration (solid blue), an economy in which attack capacity increases to $A = 5$ (dotted red), and an economy in which attacker matching efficiency increases to $\psi = 5$ (solid green). Panel B shows the same when the regulatory requirement (homogeneous across firm size with $\eta_{\text{reg}} = 0$) becomes more stringent. In the two counterfactuals, the same policy has very different implications from the benchmark economy.

Consider the subsidy policy in panel A. Unlike the benchmark economy, in which output increases up to a subsidy rate $\tau = 0.75$, when cyber agents have higher attack capacity (red dotted line), output is flat at first, and then quickly *decreases* for mid and high values of τ. This is coupled with a larger incentive to invest in cybersecurity by firms, which drives up the total tax bill much more than in the benchmark economy with $A = 1$. In contrast, with $\psi = 0.05$ the economy has output increasing for an even larger subsidy rate ($\tau = 0.9$) than the benchmark despite there too the tax to output ratio increases considerably relative to the benchmark.

The reason for the stark difference in the aggregate output response as τ changes is that, when cyber risk intensifies through higher attack capacity, subsidies disproportionately benefit large firms who are most targeted when attack capacity is high and therefore become fiscally costly, whereas when risk rises through broader attack likelihood, subsidies improve resilience across the firm distribution and remain effective. As figure 12 shows, the investment increase as the subsidy policy becomes more generous is overwhelmingly driven by the largest firms when $A = 5$. In contrast, in the $\psi = 0.05$ economy, the changes are much more homogeneous across the firm size distribution with all firms increasing their investment in cybersecurity. Therefore, unlike the $\psi = 0.05$ scenario, when $A = 5$, the subsidy acts as a net transfer from small to large firms when $A = 5$. This extra fiscal burden without a commensurate benefit, dissuades firms from entering when $A = 5$ and lowers the total mass of firms in the economy, depressing output as a consequences.

Consider the regulation case in panel B. Here too there are differences in output dynamics across regulatory thresholds and different cyber scenarios. However, even for $A = 5$, low to middle cyber requirements $\bar{r}_{\text{req}}$ still boost output. This suggests that a relatively low, homogeneous regulatory requirement might be a more robust policy than subsidies. The reason for the stark difference in the red lines of panels A and B is that, with a subsidy, small firms pay the price of the policy by effectively subsidizing large firms' cyber investments. In contrast, a modest and non-progressive regulatory require-

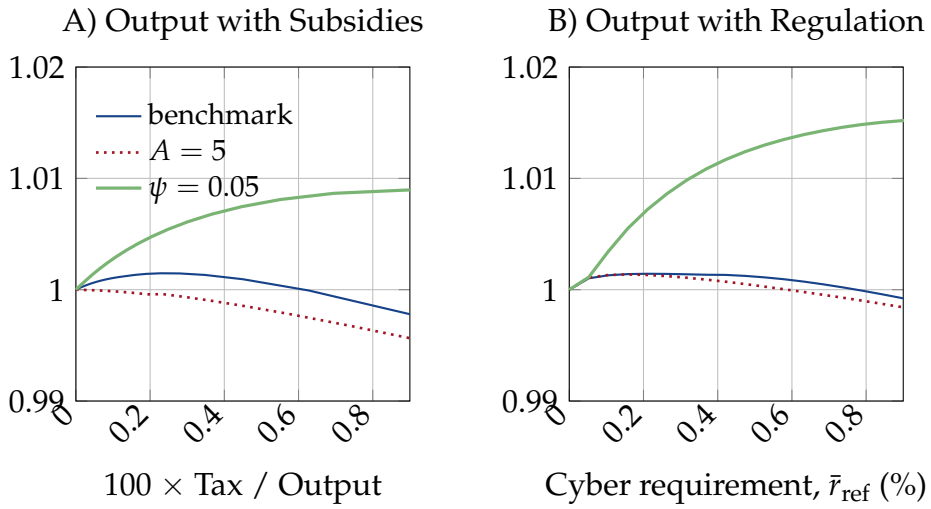


Figure 11: Output effects of subsidies and minimum regulatory requirements under alternative cyber-risk scenarios. Panel A varies subsidy generosity; Panel B varies the homogeneous minimum cyber labor requirement. Lines correspond to the benchmark, high attack capacity ($A = 5$), and high matching efficiency ($\psi = 0.05$) economies.

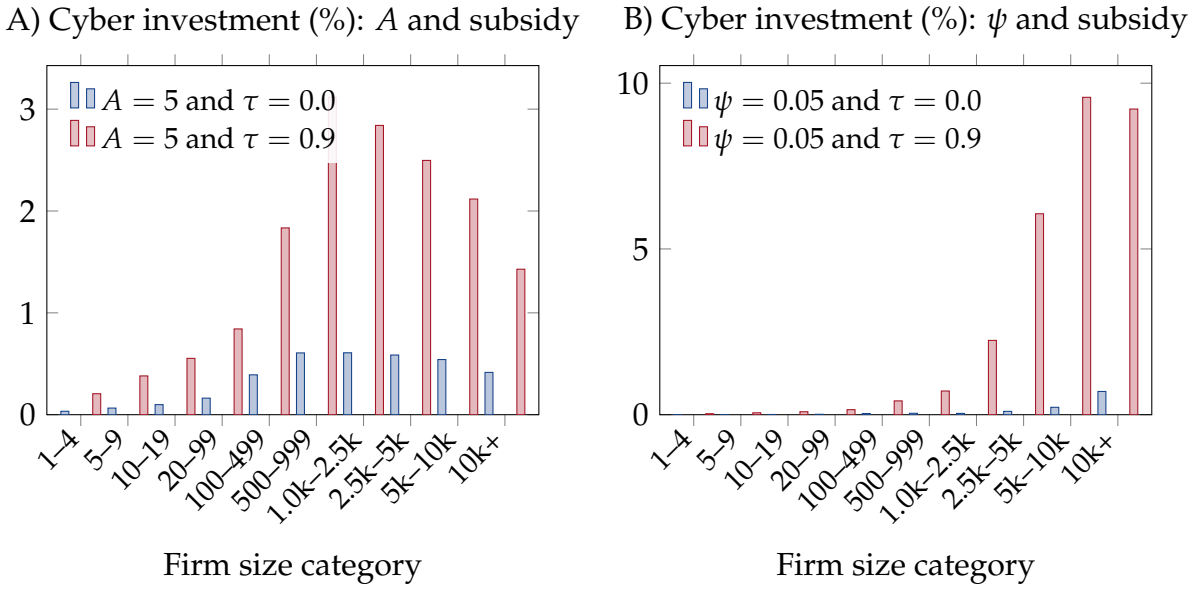


Figure 12: Cyber investment as a share of labor by firm size under adverse cyber scenarios for different subsidy policies. Panel A shows cyber investment for high attack capacity ($A = 5$) for no (blue) and high (red) subsidy rates. Panel B shows cyber investment for high attack matching efficiency ($\psi = 0.05$) for no (blue) and high (red) subsidy rates.

ments, appear to be better suited in targeting the firms that are most in need of building up their cyber defenses.

We showed earlier that attack risk is inverse-U shaped in firm size, with attacks concentrated among mid-sized firms. This suggests that an effective regulatory policy should primarily induce small and mid-sized firms to devote more labor to cybersecurity, while avoiding unnecessary distortions among large firms. Figure 13 illustrates how alternative regulatory policies affect the cyber investments of firms of different sizes. Each panel reports the cyber labor share by firm-size bin (blue bars), the minimum requirement imposed on firms in that bin (orange circles), and the cyber labor share chosen in the absence of regulation (white squares).

A low, non-progressive minimum requirement (panel A) achieves the desired reallocation well: it raises cyber labor among small and mid-sized firms relative to the unregulated economy, while remaining slack for larger firms. Introducing progressivity while keeping the baseline requirement low (panel B) raises the constraint for large firms above their unregulated cyber labor share, but the resulting distortion remains modest. By contrast, combining a higher minimum requirement with progressivity (panel D) generates a large wedge between the regulated and unregulated cyber labor shares, forcing the largest firms to devote roughly six times as much labor to cybersecurity as they would choose without regulation. This substantial distortion in labor allocation exceeds what is socially optimal.

Takeaway. In sum, this quantitative analysis highlights a sharp distinction between the policy instruments a government might use to fight cyber risk. Subsidies operate by lowering the marginal cost of preventive investment, strengthening resilience and attenuating the cyber externality, which supports entry and raises output over a wide range of policy intensities. Bailouts, by contrast, primarily provide insurance ex post, weakening investment incentives and amplifying the cyber externality, so that aggregate productivity and output decline as policy generosity increases. Regulation acts more directly by imposing minimum cybersecurity investment requirements on firms. When these requirements are moderate and well targeted, regulation can raise preventive investment among the firms most exposed to cyber risk while leaving firms that already invest sufficiently unconstrained.

Overall, the model favors a modest and homogenous regulatory requirement. The benefits of subsidies can vary substantially with the source of cyber risk, in part because subsidizing firms that would already undertake large cybersecurity investments can become fiscally costly. A moderate, homogeneous regulatory floor is more robust: it can

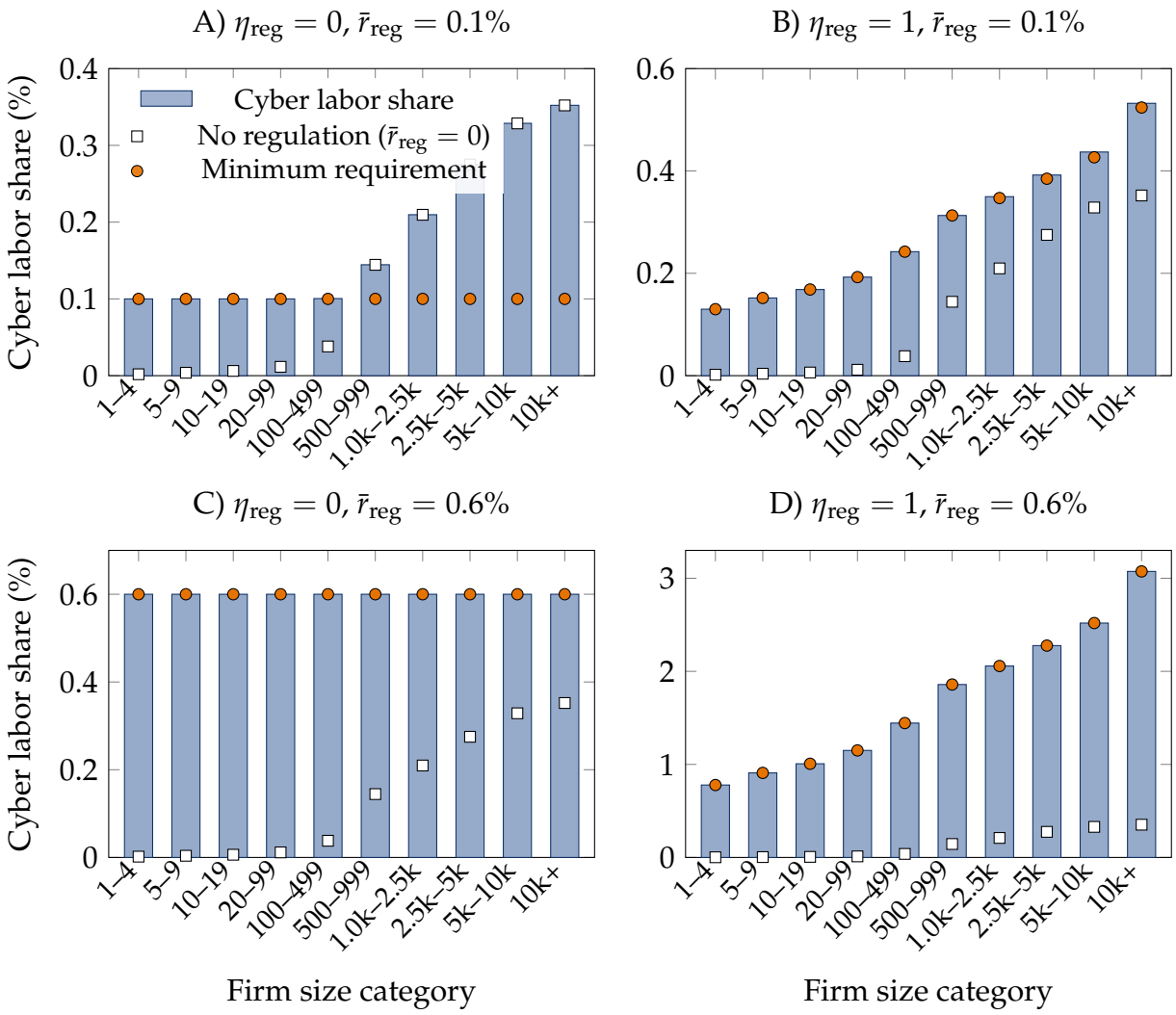


Figure 13: Each panel reports the cyber labor share by firm-size bin under regulation (blue bars), the minimum regulatory requirement for firms in that bin (orange circles), and the cyber labor share chosen in the unregulated economy (white squares). Panels vary the baseline minimum requirement ($\bar{r}_{\text{reg}}$) and the degree of progressivity (η_{reg}).

raise cybersecurity investment among the small and mid-sized firms most in need of additional protection without imposing large distortions on firms that already invest heavily. More stringent or progressive requirements, however, can instead force well-protected firms to overinvest in cybersecurity and divert labor away from production.

7 Conclusion

This paper studies cyber risk as a macroeconomic force. We develop a dynamic general equilibrium model in which heterogeneous firms invest in cybersecurity, attackers strategically target firms, and successful attacks propagate through the economy by reducing aggregate productivity. We discipline the model using new firm-level evidence showing that cybersecurity employment rises steeply with firm size.

The analysis highlights the central role of general equilibrium forces in shaping both the aggregate impact of cyber risk and the effectiveness of policy interventions. Cyberattacks generate losses not only through direct disruptions at affected firms, but also through a propagation externality that depresses economy-wide productivity and feeds back into firms' entry and investment decisions. As a result, the macroeconomic cost of cyber risk substantially exceeds the direct losses typically measured in firm-level data.

We use the model to evaluate three policy instruments that operate through distinct channels: subsidies to cybersecurity investment, minimum cyber regulatory requirements for firms, and bailouts to firms that suffer cyber losses. While partial-equilibrium intuition suggests opposing incentive effects, the general equilibrium comparison reveals a sharper contrast. Subsidies and regulatory requirements can directly strengthen preventive investment, reduce attack incidence, and mitigate the propagation externality, raising aggregate productivity, encouraging firm entry, and increasing output. Bailouts, in contrast, primarily provide ex post insurance, weakening incentives to invest through moral hazard, amplifying cyber risk, and discouraging entry. As a result, aggregate output declines as bailouts become more generous, even when fiscal costs remain limited. However, the benefits of subsidies and regulation vary depending on the nature of cyber risk, with low and homogeneous regulatory requirements providing the most robust benefits across different cyber scenarios.

Overall, the paper shows that the empirical size gradient in cybersecurity investment is a key organizing feature of cyber risk in the economy, with important implications for both aggregate outcomes and optimal policy design. Because larger firms invest more heavily in protection, attack risk is concentrated among mid-sized firms, which in turn implies these are the firms most in need of additional cybersecurity investment. This het-

erogeneity matters for intervention. Modest minimum requirements can raise investment among relatively underprotected firms while remaining slack for firms that already invest heavily. More broadly, effective cyber policy must account for how private protection incentives vary across firms and how interventions reshape entry, resource allocation, and the aggregate cyber externality in general equilibrium.

As cyber threats continue to grow in scale and sophistication, understanding their aggregate consequences and grounding policy analysis in empirically observed patterns of firm behavior will remain an increasingly important challenge for researchers and policymakers alike.

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A Model Appendix

Proof of Lemma 1.

Proof. The firm's first-order condition for cybersecurity investment is

$$w = \beta(1-\sigma)q\left[-\Lambda_x(a^*(\varphi), x^*(\varphi))\right]\left[V^s(\varphi) - V^a(\varphi)\right].$$

The term $V^s(\varphi) - V^a(\varphi)$ is the continuation-value loss from a successful attack. To see that it is increasing in φ, note that profits of a safe firm scale with productivity as $\pi(\varphi) \propto \varphi^{\frac{1}{1-\alpha}}$, while an attacked firm produces at reduced productivity $(1-\ell)\varphi$. Hence

$$\pi^a(\varphi) = \pi\big((1-\ell)\varphi\big) = (1-\ell)^{\frac{1}{1-\alpha}}\pi(\varphi),$$

so that the flow profit loss $\pi(\varphi) - \pi^a(\varphi)$ is strictly increasing in φ. Because the continuation values $V^s(\varphi)$ and $V^a(\varphi)$ inherit this scaling from current profits and future continuation payoffs, it follows that $V^s(\varphi) - V^a(\varphi)$ is strictly increasing in φ.

Define the firm's first-order condition for cybersecurity investment as

$$F(x, \varphi) \equiv w - \beta(1-\sigma)q\left[-\Lambda_x(a^*(\varphi), x)\right]\left[V^s(\varphi) - V^a(\varphi)\right] = 0.$$

Under the maintained assumptions that $C'' < 0$ and $\mathcal{A}'' < 0$, we have $\Lambda_{xx} < 0$, so that $F_x < 0$. Moreover, since $V^s(\varphi) - V^a(\varphi)$ is strictly increasing in φ, we have $F_\varphi > 0$.

Applying the implicit function theorem to $F(x^*(\varphi), \varphi) = 0$ yields

$$\frac{dx^*(\varphi)}{d\varphi} = -\frac{F_\varphi}{F_x} > 0.$$

Thus, cybersecurity investment is strictly increasing in firm productivity. □

Proof of Proposition 1.

Proof. The equilibrium probability that a matched attack succeeds is

$$\Lambda(\varphi) = \Lambda(a^*(\varphi), x^*(\varphi)) = \frac{\exp\left\{\frac{\mathcal{A}(a^*(\varphi)) - C(x^*(\varphi))}{\varrho}\right\}}{1 + \exp\left\{\frac{\mathcal{A}(a^*(\varphi)) - C(x^*(\varphi))}{\varrho}\right\}}.$$

Since Λ is strictly increasing in the net offensive advantage

$$D(\varphi) \equiv \mathcal{A}(a^*(\varphi)) - C(x^*(\varphi)),$$

it is enough to characterize the shape of $D(\varphi)$. Differentiating with respect to φ gives

$$D'(\varphi) = \mathcal{A}'(a^*(\varphi))a^{*\prime}(\varphi) - C'(x^*(\varphi))x^{*\prime}(\varphi).$$

Consider first the lower tail, $\varphi \leq \underline{\varphi}$. In this region, cybersecurity investment is only weakly increasing in productivity:

$$0 \leq x^{*\prime}(\varphi) \leq \bar{m}_x.$$

Hence the increase in effective defense is bounded above by

$$C'(x^*(\varphi))x^{*\prime}(\varphi) \leq C'(x^*(\varphi))\bar{m}_x.$$

By assumption, this term is smaller than the increase in effective attack intensity:

$$C'(x^*(\varphi))\bar{m}_x < \mathcal{A}'(a^*(\varphi))a^{*\prime}(\varphi).$$

Therefore,

$$D'(\varphi) > 0 \qquad \text{for all } \varphi \leq \underline{\varphi}.$$

Because Λ is increasing in D, it follows that

$$\Lambda'(\varphi) > 0 \qquad \text{for all } \varphi \leq \underline{\varphi}.$$

Thus, successful-attack risk rises with productivity in the lower tail.

Now consider the upper tail, $\varphi \geq \bar{\varphi}$. In this region, the attacker's capacity constraint binds, so

$$a^*(\varphi) = A \qquad \Rightarrow \qquad a^{*\prime}(\varphi) = 0.$$

Therefore,

$$D'(\varphi) = -C'(x^*(\varphi))x^{*\prime}(\varphi).$$

By Lemma 1, cybersecurity investment is increasing in productivity, so $x^{*\prime}(\varphi) > 0$. Since $C'(x) > 0$, we have

$$D'(\varphi) < 0 \qquad \text{for all } \varphi \geq \bar{\varphi}.$$

Again using the fact that Λ is increasing in D, it follows that

$$\Lambda'(\varphi) < 0 \qquad \text{for all } \varphi \geq \bar{\varphi}.$$

Combining the increasing lower tail with the decreasing upper tail, the probability of

a successful attack is hump-shaped in firm productivity. □

Proposition 1 assumes that cyber attackers have a hard capacity constraint A. In this section, we show that the same result emerges if, instead of a capacity constraint, attackers face convex costs of attack, $Q(a)$, rather than a linear cost $-a$, where costs rise fast enough with a.

Proposition 2 (Hump-shaped attack incidence with convex attack costs)**.** Suppose the attacker's payoff from choosing attack intensity a against a firm with productivity φ is

$$\Lambda(a, x^*(\varphi))W^a(\varphi) - Q(a),$$

where $Q'(a) > 0$ and $Q''(a) > 0$. Suppose the conditions of Proposition 1 governing the lower tail continue to hold, so that there exists $\underline{\varphi}$ such that

$$\Lambda'(\varphi) > 0 \qquad \text{for all } \varphi \leq \underline{\varphi}.$$

Suppose further that the attacker's solution is interior and satisfies the second-order condition, and that there exists $\bar{\varphi} > \underline{\varphi}$ such that, for all $\varphi \geq \bar{\varphi}$,

$$Q''(a^*) - \Lambda_{aa}(a^*, x^*)W^a > \frac{\mathcal{A}'(a^*)}{C'(x^*)x^{*\prime}}\left[\Lambda_a(a^*, x^*)W^{a\prime} + \Lambda_{ax}(a^*, x^*)W^a x^{*\prime}\right]. \tag{30}$$

Then the probability of a successful attack is hump-shaped in firm productivity.

Proof. Follow the proof of Proposition 1 until the upper-tail argument. In particular, letting

$$D(\varphi) \equiv \mathcal{A}(a^*(\varphi)) - C(x^*(\varphi)),$$

the conditions governing the lower tail imply

$$D'(\varphi) > 0 \qquad \text{for all } \varphi \leq \underline{\varphi},$$

and, since Λ is increasing in D,

$$\Lambda'(\varphi) > 0 \qquad \text{for all } \varphi \leq \underline{\varphi}.$$

Now consider the upper tail, $\varphi \geq \bar{\varphi}$. Rather than imposing an upper bound on attack intensity, suppose that the attacker faces the convex cost $Q(a)$. An interior optimal choice of attack intensity satisfies

$$\Lambda_a(a^*, x^*)W^a(\varphi) = Q'(a^*). \tag{31}$$

Differentiating (31) with respect to φ gives

$$\Lambda_{aa} W^a a^{*\prime} + \Lambda_{ax} W^a x^{*\prime} + \Lambda_a W^{a\prime} = Q''(a^*) a^{*\prime}.$$

Thus,

$$a^{*\prime}(\varphi) = \frac{\Lambda_a W^{a\prime} + \Lambda_{ax} W^a x^{*\prime}}{Q''(a^*) - \Lambda_{aa} W^a}. \tag{32}$$

The denominator is positive by the attacker's second-order condition.

Plugging (32) in

$$D'(\varphi) = \mathcal{A}'(a^*) a^{*\prime}(\varphi) - C'(x^*) x^{*\prime}(\varphi).$$

delivers

$$D'(\varphi) = \mathcal{A}'(a^*) \frac{\Lambda_a W^{a\prime} + \Lambda_{ax} W^a x^{*\prime}}{Q''(a^*) - \Lambda_{aa} W^a} - C'(x^*) x^{*\prime}.$$

Condition (30) therefore implies

$$D'(\varphi) < 0 \qquad \text{for all } \varphi \geq \bar{\varphi}.$$

Since Λ is increasing in D, it follows that

$$\Lambda'(\varphi) < 0 \qquad \text{for all } \varphi \geq \bar{\varphi}.$$

Combining the increasing lower tail with the decreasing upper tail, the probability of a successful attack is hump-shaped in firm productivity. □